Supporting Mathematica File for Reciprocal Preferences and Expectations in International Agreements

The number of countries is denoted by "n" instead of "N" here, as the capital N is a command in Mathematica.

Signatory and Non-signatory Efforts

Non-signatory i's best-response function

In[1]:=
```
qni[b_, c_, n_, k_, α_, η_, qqs_] :=
  (b (n - 1) + α (k qqs - (n - 1) η)) / (c (n - 1) - α (n - k - 1))
```

Signatories' objective function

In[2]:=
```
Us[b_, c_, n_, k_, α_, η_, qqs_, qqn_] :=
 b (k qqs + (n - k) qqn) - (c / 2) qqs ^ 2 + α (((k - 1) qqs + (n - k) qqn) / (n - 1) - η) (qqs + 1 - η)
```

Taking derivative of Us wrt a signatory's effort, equating it to zero, and finally solving for a signatory's effort yields

In[3]:=
```
Solve[D[Us[b, c, n, k, α, η, qqs, qni[b, c, n, k, α, η, qqs]], qqs] == 0, qqs]
```

Out[3]=

$$\left\{\left\{qqs \to \left(-\frac{b\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}-\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}-b\left(k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)+\alpha\,\eta+\frac{(-k+n)\,\alpha^2\,\eta}{c\,(-1+n)-(-1-k+n)\,\alpha}+\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)\eta}{-1+n}\right)\Big/\left(-c+\frac{(-1+k)\,\alpha}{-1+n}+\frac{k\,(-k+n)\,\alpha^2}{(-1+n)\,(c\,(-1+n)-(-1-k+n)\,\alpha)}+\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}\right)\right\}\right\}$$

Signatory effort - Equation (17):

In[4]:=

$$qs[b\_, c\_, n\_, k\_, \alpha\_, \eta\_] = \mathrm{FullSimplify}\left[\left(-\frac{b\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}-\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}-b\left(k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)+\alpha\,\eta+\frac{(-k+n)\,\alpha^2\,\eta}{c\,(-1+n)-(-1-k+n)\,\alpha}+\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)\eta}{-1+n}\right)\Big/\left(-c+\frac{(-1+k)\,\alpha}{-1+n}+\frac{k\,(-k+n)\,\alpha^2}{(-1+n)\,(c\,(-1+n)-(-1-k+n)\,\alpha)}+\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}\right)\right]$$

Out[4]=

$$\frac{b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,(\alpha-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta))}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}$$

Non-signatory effort - Equation (18):

In[5]:=
```
qn[b_, c_, n_, k_, α_, η_] = FullSimplify[qni[b, c, n, k, α, η, qs[b, c, n, k, α, η]]]
```

Out[5]=
$$\frac{b\,(-1+n) - (-1+n)\,\alpha\,\eta + \frac{k\,\alpha\,(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,(\alpha-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta)))}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}}{c\,(-1+n) + (1+k-n)\,\alpha}$$

Proposition 2

Equation (A8):

Taking derivative of signatory's effort wrt to k:

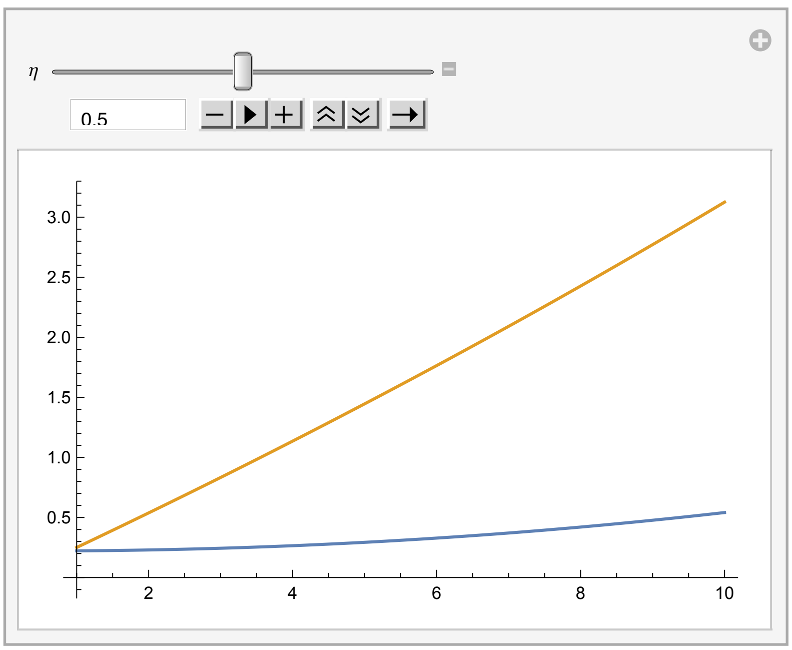
In[6]:=
```
FullSimplify[D[qs[b, c, n, k, α, η], k]]
```

Out[6]=
$$\frac{c\,(c\,(-1+n)+\alpha)\,(b\,c\,(-1+n)-b\,(-2+n)\,\alpha-\alpha\,(\alpha+c\,(-1+\eta)))}{\left(-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2\right)^2}$$

A supporting Figure of how signatory and non-signatory efforts change in k below. The formal proof is in the paper.

In[7]:=
```
Manipulate[
 Plot[{qn[0.25, 1, 10, k, 0.1, η], qs[0.25, 1, 10, k, 0.1, η]}, {k, 1, 10}], {η, 0, 1}]
```

Out[7]=



The following difference is used in the proof of proposition 2(2) as **equation (A9):**

In[8]:=
```
FullSimplify[qs[b, c, n, k, α, η] - qn[b, c, n, k, α, η]]
```

Out[8]=
$$\frac{(-1+n)\,(c\,(-1+k)+\alpha)\,(b\,c\,(-1+n)-b\,(-2+n)\,\alpha-\alpha\,(\alpha+c\,(-1+\eta)))}{(c\,(-1+n)+(1+k-n)\,\alpha)\,\left(c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2\right)}$$

Proposition 3

Notation: we use "a" for "α" lower bar to conveniently type it on Mathematica.
Let's start with **the signatory effort:**

In[9]:=
```
qsa[b_, c_, n_, k_, a_, η_] := qs[b, c, n, k, (a c)/(c - b), η]
```

Taking derivative of signatory effort wrt a:

In[10]:=
```
Dqsaa[b_, c_, n_, k_, a_, η_] = FullSimplify[D[qsa[b, c, n, k, a, η], a]]
```

Out[10]=
$$-\left(\left((b-c)\left(a^2\,(2\,b\,n+c\,(1+k-n-2\,\eta))-2\,a\,(b-c)\,(-1+n)\,(c+2\,b\,k-2\,c\,\eta)+(b-c)^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))\right)\right)\Big/\left(c\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2\right)\right)$$

In[11]:=
$$-\left(\left((b-c)\left(a^2\,(2\,b\,n+c\,(1+k-n-2\,\eta))-2\,a\,(b-c)\,(-1+n)\,(c+2\,b\,k-2\,c\,\eta)+(b-c)^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))\right)\right)\Big/\left(c\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2\right)\right)$$

Out[11]=
$$-\left(\left((b-c)\left(a^2\,(2\,b\,n+c\,(1+k-n-2\,\eta))-2\,a\,(b-c)\,(-1+n)\,(c+2\,b\,k-2\,c\,\eta)+(b-c)^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))\right)\right)\Big/\left(c\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2\right)\right)$$

Evaluating at a=0:

In[12]:=
```
FullSimplify[Dqsaa[b, c, n, k, 0, η]]
```

Out[12]=
$$-\frac{b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta)}{(b-c)\,c\,(-1+n)}$$

Solving for the critical eta threshold, yields **T_H:**

In[13]:=
```
FullSimplify[Solve[b (n + k (-3 + k + n)) + c (-1 + k - (-2 + k + n) η) == 0, η]]
```

Out[13]=
$$\left\{\left\{\eta\to\frac{c\,(-1+k)+b\,(n+k\,(-3+k+n))}{c\,(-2+k+n)}\right\}\right\}$$

Non-signatory effort:

In[14]:=
```
qna[b_, c_, n_, k_, a_, η_] := FullSimplify[qn[b, c, n, k, (a c)/(c - b), η]]
```

Derivative wrt a:

In[15]:= Dqnaa[b_, c_, n_, k_, a_, η_] = FullSimplify[D[qna[b, c, n, k, a, η], a]]

Out[15]=
$$\frac{1}{c\left(-1+\frac{a(1+k-n)}{-b+c}+n\right)^2}\left(\frac{1}{b-c}\left(-1+\frac{a(1+k-n)}{-b+c}+n\right)\left(c(-1+n)\eta+\frac{a k(2 a c(-1+2\eta)+(b-c)(b n+c(-1+k-(-2+k+n)\eta)))}{-2a^2+(b-c)^2(-1+n)+a(b-c)(-3+k+n)}-\left(a k(-4a+(b-c)(-3+k+n))\left(-b(b-c)^2 k(-1+n)+a^2 c(-1+2\eta)+a(b-c)(b n+c(-1+k-(-2+k+n)\eta))\right)\right)\Big/\left(-2a^2+(b-c)^2(-1+n)+a(b-c)(-3+k+n)\right)^2+\frac{k\left(-b(b-c)^2 k(-1+n)+a^2 c(-1+2\eta)+a(b-c)(b n+c(-1+k-(-2+k+n)\eta))\right)}{-2a^2+(b-c)^2(-1+n)+a(b-c)(-3+k+n)}\right)-\frac{(1+k-n)\left(b(-1+n)+\frac{a c(-1+n)\eta}{b-c}+\frac{a k(-b(b-c)^2 k(-1+n)+a^2 c(-1+2\eta)+a(b-c)(b n+c(-1+k-(-2+k+n)\eta)))}{(b-c)(-2a^2+(b-c)^2(-1+n)+a(b-c)(-3+k+n))}\right)}{-b+c}\right)$$

Evaluating at a=0:

In[16]:= FullSimplify[Dqnaa[b, c, n, k, 0, η]]

Out[16]=
$$\frac{b(-1+(-1+k)k+n)-c(-1+n)\eta}{c(-b+c)(-1+n)}$$

Solving for the critical eta threshold, yields **T_L:**

In[17]:= FullSimplify[Solve[$\frac{b(-1+(-1+k)k+n)-c(-1+n)\eta}{c(-b+c)(-1+n)}$ == 0, η]]

Out[17]=
$$\left\{\left\{\eta\to\frac{b(-1+(-1+k)k+n)}{c(-1+n)}\right\}\right\}$$

T_H - T_L:

In[18]:= diff[b_, c_, k_, n_] =
FullSimplify[$\frac{c(-1+k)+b(n+k(-3+k+n))}{c(-2+k+n)}-\frac{b(-1+(-1+k)k+n)}{c(-1+n)}$]

Out[18]=
$$\frac{b k-\frac{b(-1+k)k}{-1+n}-\frac{(2b-c)(-1+k)}{-2+k+n}}{c}$$

For the boundary of T_H:

In[19]:= FullSimplify[Solve[k (n + k - 3) - n (n - 2) == 0, k]]

Out[19]=
$$\left\{\left\{k\to\frac{1}{2}\left(3-n-\sqrt{(-1+n)(-9+5n)}\right)\right\},\left\{k\to\frac{1}{2}\left(3-n+\sqrt{(-1+n)(-9+5n)}\right)\right\}\right\}$$

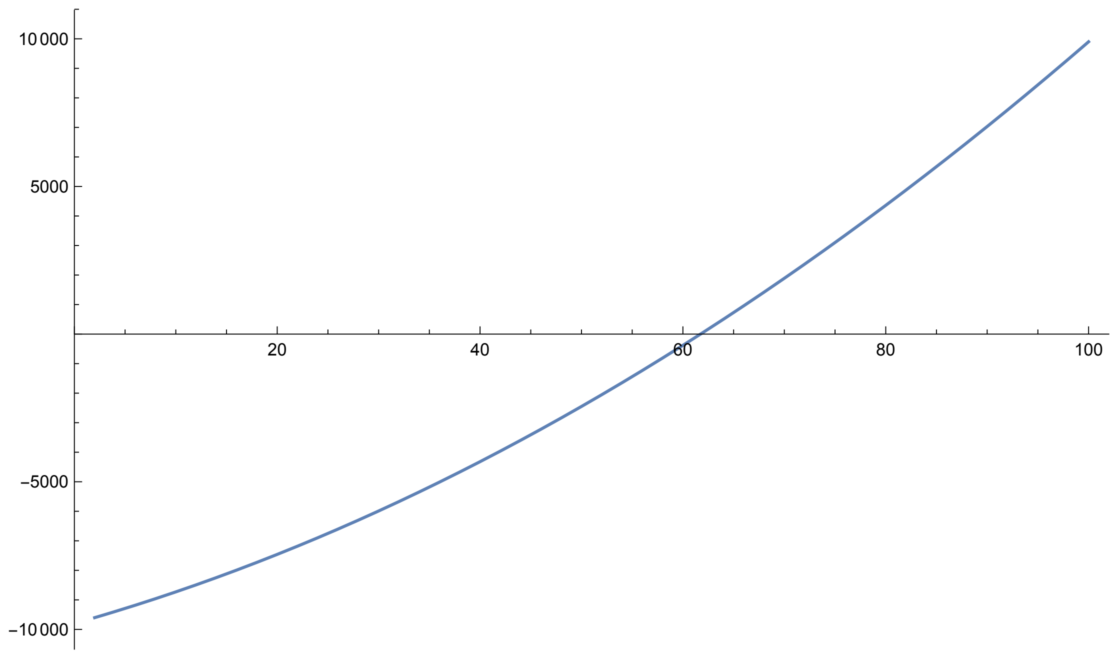
For n=100, the smallest integer above the higher of the two numbers above shows for k > 62, we have T_H >1:

In[20]:= Ceiling[$\frac{1}{2}\left(3-100+\sqrt{(-1+100)(-9+5\times100)}\right)$]

Out[20]= 62

In[21]:=
```
Plot[k (100 + k - 3) - 100 (100 - 2), {k, 2, 100}]
```

Out[21]=



Indirect Utility Functions with alpha

We omit writing these indirect utility functions omitted in the paper due to their size and complexity.
Signatory

In[22]:=
```
FullSimplify[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]]]
```

Out[22]=
$$\frac{1}{2\,(c\,(-1+n)+(1+k-n)\,\alpha)\,\left(c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2\right)}\Big(b^2\,\big(c^2\,(-1+n)^2\,((-2+k)\,k+2\,n)+2\,c\,(-1+n)\,\left(k^2+k\,(-3+n)-(-3+n)\,n\right)\alpha+(4\,k\,(-1+n)+(4-3\,n)\,n)\,\alpha^2\big)+2\,b\,\alpha\,\Big((c\,(-1+n)+\alpha)\,(c\,((-2+k)\,k+n)+(2\,k-n)\,\alpha)+\left(-c^2\,(-1+n)\,\left((-2+k)\,k+n^2\right)+c\,\left(2\,k-k^2\,n+n\,(2+(-4+n)\,n)\right)\alpha+2\,n\,(-1-k+n)\,\alpha^2\right)\eta\Big)+\alpha\,\Big(\alpha^3+2\,c^3\,(-1+n)^2\,(-1+\eta)\,\eta+2\,c\,\alpha^2\,\left(-1+k\,(-1+\eta)^2+\eta\,(-1+2\,n+\eta-n\,\eta)\right)+c^2\,\alpha\,\left(1-2\,k\,(-1+\eta)^2+k^2\,(-1+\eta)^2+\eta\,(4+2\,(-4+n)\,n-(4+(-6+n)\,n)\,\eta)\right)\Big)\Big)$$

Non-signatory

In[23]:=
```
Un[b_, c_, n_, k_, α_, η_, qqs_, qqn_] =
 b (k qqs + (n - k) qqn) - (c / 2) qqn ^ 2 + α ((k qqs + (n - k - 1) qqn) / (n - 1) - η ) (qqn + 1 - η)
```

Out[23]=
$$-\frac{c\,qqn^2}{2}+b\,((-k+n)\,qqn+k\,qqs)+\alpha\,(1+qqn-\eta)\,\left(\frac{(-1-k+n)\,qqn+k\,qqs}{-1+n}-\eta\right)$$

In[24]:= `FullSimplify[Un[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]]]`

Out[24]=
$$\left(\alpha\left(b\left(c^2(-1+n)^2+c(3+(-1+k)k-n)(-1+n)\alpha+(2+(-2+k)n)\alpha^2\right)-\left(c^2(-1+n)^2-c(-3+n)(-1+n)\alpha+(2+k-2n)\alpha^2\right)(\alpha+c(-1+\eta))\right)\left(b\left(c^2(-1+n)(-1+(-1+k)k+n)+c\left(-3-2k+k^2+(4+k)n-n^2\right)\alpha+2(1+k-n)\alpha^2\right)+c\left(-c^2(-1+n)^2\eta+\alpha^2(k-2(1+k-n)\eta)+c\alpha\left((-1+k)k+\left(3+k-k^2-4n+n^2\right)\eta\right)\right)\right)\right)\Big/\left((c(-1+n)+(1+k-n)\alpha)^2\left(-c^2(-1+n)+c(-3+k+n)\alpha+2\alpha^2\right)^2\right)-\frac{c\left(b(-1+n)-(-1+n)\alpha\eta+\frac{k\alpha(bck(-1+n)+bn\alpha+\alpha(\alpha-2\alpha\eta+c(-1+k-(-2+k+n)\eta)))}{c^2(-1+n)-c(-3+k+n)\alpha-2\alpha^2}\right)^2}{2(c(-1+n)+(1+k-n)\alpha)^2}+b\left(\frac{k(bck(-1+n)+bn\alpha+\alpha(\alpha-2\alpha\eta+c(-1+k-(-2+k+n)\eta)))}{c^2(-1+n)-c(-3+k+n)\alpha-2\alpha^2}+\frac{(-k+n)\left(b(-1+n)-(-1+n)\alpha\eta+\frac{k\alpha(bck(-1+n)+bn\alpha+\alpha(\alpha-2\alpha\eta+c(-1+k-(-2+k+n)\eta)))}{c^2(-1+n)-c(-3+k+n)\alpha-2\alpha^2}\right)}{c(-1+n)+(1+k-n)\alpha}\right)$$

Self-interested non-signatory and signatory:

In[25]:=
```
SelfUn[b_, c_, n_, k_] =
 FullSimplify[Un[b, c, n, k, 0, η, qs[b, c, n, k, 0, η], qn[b, c, n, k, 0, η]]]
SelfUs[b_, c_, n_, k_] = FullSimplify[
  Us[b, c, n, k, 0, η, qs[b, c, n, k, 0, η], qn[b, c, n, k, 0, η]]]
```

Out[25]= $$\frac{b^2(-1+2(-1+k)k+2n)}{2c}$$

Out[26]= $$\frac{b^2((-2+k)k+2n)}{2c}$$

Stable coalition size for self-interested:

In[27]:= `FullSimplify[Solve[SelfUs[b, c, n, k] - SelfUn[b, c, n, k - 1] == 0, k]]`

Out[27]= $\{\{k\to 1\},\{k\to 3\}\}$

Lemma 1

The partial derivative of signatory indirect welfare with respect to k:

In[28]:= `FullSimplify[D[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]], k]]`

Out[28]= $$\frac{(c(-1+k)+\alpha)(c(-1+n)+\alpha)^2(b(c-cn+(-2+n)\alpha)+\alpha(\alpha+c(-1+\eta)))^2}{(c(-1+n)+(1+k-n)\alpha)^2\left(-c^2(-1+n)+c(-3+k+n)\alpha+2\alpha^2\right)^2}$$

Coalition size minimizing signatory's indirect welfare:

In[29]:=
```
FullSimplify[
 Solve[D[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]], k] == 0, k]]
```

Out[29]= $\left\{\left\{k\to 1-\frac{\alpha}{c}\right\}\right\}$

are used in the **proof of parts (1 and 2).**
At what k signatory and non-signatory indirect welfare are equal?

In[30]:= FullSimplify[Solve[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]] - Un[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]] == 0, k]]

Out[30]= $\left\{\left\{k \to 1 - \frac{\alpha}{c}\right\}, \left\{k \to -\frac{(-1+n)\,(c-\alpha)\,(c\,(-1+n)+2\,\alpha)}{c^2\,(-1+n)^2 + 2\,c\,(-1+n)\,\alpha + 2\,\alpha^2}\right\}\right\}$

The latter is negative. So, they are equal at k = 1 - $\frac{\alpha}{c}$. **This proves the part 4 and it is useful to prove the parts 3 and 5.**

Next, we study **(Un - Us)/(k-1 + α/c):**

In[31]:= FullSimplify[(Un[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]] - Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]]) / (k - 1 + α / c)]

Out[31]= $\left(c\left(c^2\,(1+k)\,(-1+n)^2 + c\,(3+2\,k-n)\,(-1+n)\,\alpha + 2\,(1+k-n)\,\alpha^2\right)\left(b\,(c-c\,n+(-2+n)\,\alpha)+\alpha\,(\alpha+c\,(-1+\eta))\right)^2\right) \Big/ \left(2\,(c\,(-1+n)+(1+k-n)\,\alpha)^2\left(-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2\right)^2\right)$

For this term to be positive, the first term in the numerator has to be positive since the rest of the terms are non-zero and squared.

In[32]:= termunus[c_, n_, k_, α_] = $c^2\,(1+k)\,(-1+n)^2 + c\,(3+2\,k-n)\,(-1+n)\,\alpha + 2\,(1+k-n)\,\alpha^2$

Out[32]= $c^2\,(1+k)\,(-1+n)^2 + c\,(3+2\,k-n)\,(-1+n)\,\alpha + 2\,(1+k-n)\,\alpha^2$

Does this term increase in c?

In[33]:= FullSimplify[D[termunus[c, n, k, α], c]]

Out[33]= $(-1+n)\,(2\,c\,(1+k)\,(-1+n) + (3+2\,k-n)\,\alpha)$

In[34]:= FullSimplify[(-1 + n) (4 α (1 + k) (-1 + n) + (3 + 2 k - n) α)]

Out[34]= $(-1+n)\,(-1+3\,n+k\,(-2+4\,n))\,\alpha$

In[35]:= FullSimplify[termunus[2 α, n, k, α]]

Out[35]= $2\,(k+(-1+n)\,n+2\,k\,(-1+n)\,n)\,\alpha^2$

Lemma 2

Let us start with the U_s.
The partial derivative of signatory indirect utility wrt qn

In[36]:= $D\left[Us\left[b, c, n, k, \frac{a\,c}{c-b}, \eta, qs, qn\right], qn\right]$

Out[36]= $b\,(-k+n) + \frac{a\,c\,(-k+n)\,(1+qs-\eta)}{(-b+c)\,(-1+n)}$

The partial derivative of non-signatory effort wrt a

In[37]:=
```
FullSimplify[D[qn[b, c, n, k, a c/(c - b), η], a]]
```

Out[37]=

$$\frac{1}{(b-c)\,c\left(-1+\frac{a(1+k-n)}{-b+c}+n\right)^2\left(-2a^2+(b-c)^2(-1+n)+a(b-c)(-3+k+n)\right)^2}\Big(a^4\left(-2b(1+k-n)(2+(-2+k)n)-ck\left(3+2k+k^2-2(2+k)n+n^2\right)+2c(2+k-2n)(1+k-n)\eta\right)+4a^3(b-c)(-1+n)\left(-ck(-2+n)+b\left(3+2k+k^3-\left(4+k^2\right)n+n^2\right)+c(1+k-n)(-3+n)\eta\right)-(b-c)^4(-1+n)^3(b(-1+(-1+k)k+n)-c(-1+n)\eta)-a^2(b-c)^2(-1+n)\left(ck(-7-2k(-2+n)+5n)+b\left(-13-7k+(-3+k)k^3+23n+k(6+5k)n-\left(11+k+k^2\right)n^2+n^3\right)+c\left(k(8-6n)+k^2(-3+n)-(-1+n)(13+(-10+n)n)\right)\eta\right)+2a(b-c)^3(-1+n)^2\left(b\left(-3-2k+k^2+(4+k)n-n^2\right)+c\left((-1+k)k+\left(3+k-k^2-4n+n^2\right)\eta\right)\right)\Big)$$

The partial derivative of signatory indirect utility wrt qn (qs at optimal) times the partial derivative of non-signatory effort wrt a

In[38]:=
```
derUsqna[b_, c_, n_, k_, a_, η_] :=
 FullSimplify[(b (-k + n) + (a c (-k + n) (1 + qs[b, c, n, k, a c/(c-b), η] - η))/((-b + c) (-1 + n)))
  ((a^4 (-2 b (1 + k - n) (2 + (-2 + k) n) - c k (3 + 2 k + k^2 - 2 (2 + k) n + n^2) +
        2 c (2 + k - 2 n) (1 + k - n) η) + 4 a^3 (b - c) (-1 + n)
       (-c k (-2 + n) + b (3 + 2 k + k^3 - (4 + k^2) n + n^2) + c (1 + k - n) (-3 + n) η) -
      (b - c)^4 (-1 + n)^3 (b (-1 + (-1 + k) k + n) - c (-1 + n) η) -
      a^2 (b - c)^2 (-1 + n) (c k (-7 - 2 k (-2 + n) + 5 n) +
         b (-13 - 7 k + (-3 + k) k^3 + 23 n + k (6 + 5 k) n - (11 + k + k^2) n^2 + n^3) +
         c (k (8 - 6 n) + k^2 (-3 + n) - (-1 + n) (13 + (-10 + n) n)) η) + 2 a (b - c)^3 (-1 + n)^2
       (b (-3 - 2 k + k^2 + (4 + k) n - n^2) + c ((-1 + k) k + (3 + k - k^2 - 4 n + n^2) η))) /
    ((b - c) c (-1 + (a (1 + k - n))/(-b + c) + n)^2 (-2 a^2 + (b - c)^2 (-1 + n) + a (b - c) (-3 + k + n))^2))]
```

Evaluating at alpha = 0.

In[39]:=
```
FullSimplify[derUsqna[b, c, n, k, 0, η]]
```

Out[39]=

$$\frac{b(k-n)(b(-1+(-1+k)k+n)-c(-1+n)\eta)}{(b-c)\,c\,(-1+n)}$$

The partial derivative of signatory indirect utility wrt qs

In[40]:=
```
D[Us[b, c, n, k, a c/(c - b), η, qs, qn], qs]
```

Out[40]=

$$b\,k-c\,qs+\frac{a\,c(-1+k)(1+qs-\eta)}{(-b+c)(-1+n)}+\frac{a\,c\left(\frac{(-k+n)qn+(-1+k)qs}{-1+n}-\eta\right)}{-b+c}$$

The partial derivative of signatory effort wrt a

In[41]:=
$$\text{FullSimplify}\left[\text{D}\left[\text{qs}\left[\text{b, c, n, k, }\frac{\text{a c}}{\text{c}-\text{b}}\text{, }\eta\right]\text{, a}\right]\right]$$

Out[41]=
$$-\left(\left((b-c)\left(a^2\,(2\,b\,n+c\,(1+k-n-2\,\eta))-2\,a\,(b-c)\,(-1+n)\,(c+2\,b\,k-2\,c\,\eta)+(b-c)^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))\right)\right)\Big/\left(c\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2\right)\right)$$

The partial derivative of signatory indirect utility wrt qs (qs and qn at optimal) times t partial derivative of signatory effort wrt a

In[42]:=
$$\text{derUsqsa[b\_, c\_, n\_, k\_, a\_, }\eta\text{\_] := FullSimplify}\Bigg[\left(b\,k-c\left(\text{qs}\left[b,c,n,k,\frac{a\,c}{c-b},\eta\right]\right)+\frac{a\,c\,(-1+k)\left(1+\left(\text{qs}\left[b,c,n,k,\frac{a\,c}{c-b},\eta\right]\right)-\eta\right)}{(-b+c)\,(-1+n)}+\frac{a\,c\left(\frac{(-k+n)\left(\text{qn}\left[b,c,n,k,\frac{a\,c}{c-b},\eta\right]\right)+(-1+k)\left(\text{qs}\left[b,c,n,k,\frac{a\,c}{c-b},\eta\right]\right)}{-1+n}-\eta\right)}{-b+c}\right)$$
$$\left(-\left(\left((b-c)\left(a^2\,(2\,b\,n+c\,(1+k-n-2\,\eta))-2\,a\,(b-c)\,(-1+n)\,(c+2\,b\,k-2\,c\,\eta)+(b-c)^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))\right)\right)\Big/\left(c\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2\right)\right)\right)\Bigg]$$

In[43]:= `FullSimplify[derUsqsa[b, c, n, k, 0, η]]`

Out[43]= 0

Combining both:

In[44]:= `derUsazero[b_, c_, n_, k_, η_] = FullSimplify[derUsqna[b, c, n, k, 0, η] + derUsqsa[b, c, n, k, 0, η]]`

Out[44]=
$$\frac{b\,(k-n)\,(b\,(-1+(-1+k)\,k+n)-c\,(-1+n)\,\eta)}{(b-c)\,c\,(-1+n)}$$

Finding the critical level of eta that equate the term above to zero:

In[45]:= `FullSimplify[Solve[derUsazero[b, c, n, k, η] == 0, η]]`

Out[45]=
$$\left\{\left\{\eta\to\frac{b\,(-1+(-1+k)\,k+n)}{c\,(-1+n)}\right\}\right\}$$

which is identical to T_L and completes the **part 1, see equation (A12) in the paper.**

Let us continue with the U_n.
The partial derivative of non-signatory indirect utility wrt qs

In[46]:=
$$\text{D}\left[\text{Un}\left[\text{b, c, n, k, }\frac{\text{a c}}{\text{c}-\text{b}}\text{, }\eta\text{, qs, qn}\right]\text{, qs}\right]$$

Out[46]=
$$b\,k+\frac{a\,c\,k\,(1+qn-\eta)}{(-b+c)\,(-1+n)}$$

The partial derivative of signatory effort wrt a

In[47]:=
$$\text{FullSimplify}\left[D\left[qs\left[b, c, n, k, \frac{a\,c}{c-b}, \eta\right], a\right]\right]$$

Out[47]=
$$-\left(\left((b-c)\left(a^2\,(2\,b\,n+c\,(1+k-n-2\,\eta))-2\,a\,(b-c)\,(-1+n)\,(c+2\,b\,k-2\,c\,\eta)+(b-c)^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))\right)\right)\Big/\left(c\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2\right)\right)$$

The partial derivative of non-signatory indirect utility wrt qs (qn at optimal) times the partial derivative of signatory effort wrt alpha

In[48]:=
$$\text{derUnqsa}[b\_, c\_, n\_, k\_, a\_, \eta\_] := \text{FullSimplify}\Bigg[\left(b\,k+\frac{a\,c\,k\left(1+\left(qn\left[b, c, n, k, \frac{a\,c}{c-b}, \eta\right]\right)-\eta\right)}{(-b+c)\,(-1+n)}\right)\left(-\left(\left((b-c)\left(a^2\,(2\,b\,n+c\,(1+k-n-2\,\eta))-2\,a\,(b-c)\,(-1+n)\,(c+2\,b\,k-2\,c\,\eta)+(b-c)^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))\right)\right)\Big/\left(c\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2\right)\right)\right)\Bigg]$$

Evaluating at alpha = 0.

In[49]:=
$$\text{FullSimplify}[\text{derUnqsa}[b, c, n, k, 0, \eta]]$$

Out[49]=
$$-\frac{b\,k\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))}{(b-c)\,c\,(-1+n)}$$

The partial derivative of non-signatory indirect utility wrt qn

In[50]:=
$$\text{FullSimplify}\left[D\left[Un\left[b, c, n, k, \frac{a\,c}{c-b}, \eta, qs, qn\right], qn\right]\right]$$

Out[50]=
$$b\,(-k+n)-c\,qn+\frac{a\,c\,(-(-1+n)\,(1+2\,qn-2\,\eta)+k\,(1+2\,qn-qs-\eta))}{(b-c)\,(-1+n)}$$

The partial derivative of non-signatory effort wrt a

In[51]:=
$$\text{FullSimplify}\left[D\left[qn\left[b, c, n, k, \frac{a\,c}{c-b}, \eta\right], a\right]\right]$$

Out[51]=
$$\frac{1}{(b-c)\,c\left(-1+\frac{a\,(1+k-n)}{-b+c}+n\right)^2\left(-2\,a^2+(b-c)^2\,(-1+n)+a\,(b-c)\,(-3+k+n)\right)^2}\Big(a^4\left(-2\,b\,(1+k-n)\,(2+(-2+k)\,n)-c\,k\left(3+2\,k+k^2-2\,(2+k)\,n+n^2\right)+2\,c\,(2+k-2\,n)\,(1+k-n)\,\eta\right)+4\,a^3\,(b-c)\,(-1+n)\left(-c\,k\,(-2+n)+b\left(3+2\,k+k^3-\left(4+k^2\right)n+n^2\right)+c\,(1+k-n)\,(-3+n)\,\eta\right)-(b-c)^4\,(-1+n)^3\,(b\,(-1+(-1+k)\,k+n)-c\,(-1+n)\,\eta)-a^2\,(b-c)^2\,(-1+n)\left(c\,k\,(-7-2\,k\,(-2+n)+5\,n)+b\left(-13-7\,k+(-3+k)\,k^3+23\,n+k\,(6+5\,k)\,n-\left(11+k+k^2\right)n^2+n^3\right)+c\left(k\,(8-6\,n)+k^2\,(-3+n)-(-1+n)\,(13+(-10+n)\,n)\right)\eta\right)+2\,a\,(b-c)^3\,(-1+n)^2\left(b\left(-3-2\,k+k^2+(4+k)\,n-n^2\right)+c\left((-1+k)\,k+\left(3+k-k^2-4\,n+n^2\right)\eta\right)\right)\Big)$$

The partial derivative of non-signatory indirect utility wrt qn times the partial derivative of non-signatory effort wrt a

In[52]:=
```
derUnqna[b_, c_, n_, k_, a_, η_] :=
 FullSimplify[(b (-k + n) - c (qn[b, c, n, k, a c/(c - b), η]) +
      1/((b - c) (-1 + n)) a c (-(-1 + n) (1 + 2 (qn[b, c, n, k, a c/(c - b), η]) - 2 η) +
         k (1 + 2 (qn[b, c, n, k, a c/(c - b), η]) - (qs[b, c, n, k, a c/(c - b), η]) - η)))
   ((a^4 (-2 b (1 + k - n) (2 + (-2 + k) n) - c k (3 + 2 k + k^2 - 2 (2 + k) n + n^2) +
         2 c (2 + k - 2 n) (1 + k - n) η) + 4 a^3 (b - c) (-1 + n)
       (-c k (-2 + n) + b (3 + 2 k + k^3 - (4 + k^2) n + n^2) + c (1 + k - n) (-3 + n) η) -
      (b - c)^4 (-1 + n)^3 (b (-1 + (-1 + k) k + n) - c (-1 + n) η) -
      a^2 (b - c)^2 (-1 + n) (c k (-7 - 2 k (-2 + n) + 5 n) +
         b (-13 - 7 k + (-3 + k) k^3 + 23 n + k (6 + 5 k) n - (11 + k + k^2) n^2 + n^3) +
         c (k (8 - 6 n) + k^2 (-3 + n) - (-1 + n) (13 + (-10 + n) n)) η) + 2 a (b - c)^3 (-1 + n)^2
       (b (-3 - 2 k + k^2 + (4 + k) n - n^2) + c ((-1 + k) k + (3 + k - k^2 - 4 n + n^2) η)))/
    ((b - c) c (-1 + (a (1 + k - n))/(-b + c) + n)^2 (-2 a^2 + (b - c)^2 (-1 + n) + a (b - c) (-3 + k + n))^2))]
```

Evaluating at alpha = 0.

In[53]:=
```
FullSimplify[derUnqna[b, c, n, k, 0, η]]
```

Out[53]=
$$\frac{b\,(1+k-n)\,(b\,(-1+(-1+k)\,k+n)-c\,(-1+n)\,\eta)}{(b-c)\,c\,(-1+n)}$$

Combining both:

In[54]:=
```
derUnazero[b_, c_, n_, k_, η_] =
 FullSimplify[derUnqsa[b, c, n, k, 0, η] + derUnqna[b, c, n, k, 0, η]]
```

Out[54]=
$$\frac{b\left(-b\left(-k\,(-2+n)+(-1+n)^2+k^2\,(-3+2\,n)\right)+c\left(k+k^2\,(-1+\eta)-k\,\eta+(-1+n)^2\,\eta\right)\right)}{(b-c)\,c\,(-1+n)}$$

In[55]:=
```
FullSimplify[Solve[derUnazero[b, c, n, k, η] == 0, η]]
```

Out[55]=
$$\left\{\left\{\eta\to\frac{c\,(-1+k)\,k+b\left(-k\,(-2+n)+(-1+n)^2+k^2\,(-3+2\,n)\right)}{c\left((-1+k)\,k+(-1+n)^2\right)}\right\}\right\}$$

which we call T_{U_{ns}} in the paper in **equation (A13)**, completing part 2.

The partial derivative of non-signatory indirect welfare wrt a at k-1 and evaluating at a = 0:

In[56]:=
```
FullSimplify[derUnazero[b, c, n, k - 1, η]]
```

Out[56]=
$$\frac{1}{(b-c)\,c\,(-1+n)}b\left(-c\,(-2+k)\,(-1+k)-b\left(-4+(8-3\,k)\,k+n+k\,(-5+2\,k)\,n+n^2\right)+c\,(3+(-3+k)\,k+(-2+n)\,n)\,\eta\right)$$

In[57]:=
```
FullSimplify[Solve[derUnazero[b, c, n, k - 1, η] == 0, η]]
```

Out[57]=
$$\left\{\left\{\eta\to\frac{c\,(-2+k)\,(-1+k)+b\left(-4+(8-3\,k)\,k+n+k\,(-5+2\,k)\,n+n^2\right)}{c\,(3+(-3+k)\,k+(-2+n)\,n)}\right\}\right\}$$

which we call T_{U_{ns}(k-1)} in the paper in **equation (A14)**, completing part 3.

Let us compare these three thresholds, starting with **T_Uns -T_L:**

In[58]:= $$\text{FullSimplify}\left[\frac{c\,(-1+k)\,k+b\left(-k\,(-2+n)+(-1+n)^2+k^2\,(-3+2\,n)\right)}{c\left((-1+k)\,k+(-1+n)^2\right)}-\frac{b\,(-1+(-1+k)\,k+n)}{c\,(-1+n)}\right]$$

Out[58]= $$\frac{k\,(2\,b\,(-1+n)+c\,(-1+k)\,(-1+n)+b\,k\,(2-(-2+k)\,k+(-4+n)\,n))}{c\left((-1+k)\,k+(-1+n)^2\right)\,(-1+n)}$$

T_Uns(k) - T_Uns(k-1):

In[59]:= $$\text{FullSimplify}\left[\frac{c\,(-1+k)\,k+b\left(-k\,(-2+n)+(-1+n)^2+k^2\,(-3+2\,n)\right)}{c\left((-1+k)\,k+(-1+n)^2\right)}-\frac{c\,(-2+k)\,(-1+k)+b\left(-4+(8-3\,k)\,k+n+k\,(-5+2\,k)\,n+n^2\right)}{c\,(3+(-3+k)\,k+(-2+n)\,n)}\right]$$

Out[59]= $$\frac{(-1+n)\left(2\,c\,(-1+k)\,(-1+n)-b\left(k^2-k\,(3-2\,n)^2+(-1+n)\,(-7+3\,n)\right)\right)}{c\left((-1+k)\,k+(-1+n)^2\right)\,(3+(-3+k)\,k+(-2+n)\,n)}$$

For this term to be positive, we need $2\,c\left(-1+k\right)\,(-1+n)-b\left(k^2-k\left(3-2\,n\right)^2+(-1+n)\,\left(-7+3\,n\right)\right)>0$. Taking the term starting with -b to the RHS, leaving b/c on the RHS, multiplying both sides with n yields the following since bn/c=1:

In[60]:= $$\text{FullSimplify}\left[2\,(-1+k)\,(-1+n)-\left(k^2-k\,(3-2\,n)^2+(-1+n)\,(-7+3\,n)\right)>0\right]$$

Out[60]= $$(8-3\,n)\,n+k\,(7+2\,n\,(-5+2\,n))>5+k^2$$

T_L - T_Uns(k-1):

In[61]:= $$\text{FullSimplify}\left[\frac{b\,(-1+(-1+k)\,k+n)}{c\,(-1+n)}-\frac{c\,(-2+k)\,(-1+k)+b\left(-4+(8-3\,k)\,k+n+k\,(-5+2\,k)\,n+n^2\right)}{c\,(3+(-3+k)\,k+(-2+n)\,n)}\right]$$

Out[61]= $$\frac{(-1+k)\left(-c\,(-2+k)\,(-1+n)+b\left(7+k\,(-1+(-3+k)\,k)-10\,n+4\,k\,n-(-3+k)\,n^2\right)\right)}{c\,(-1+n)\,(3+(-3+k)\,k+(-2+n)\,n)}$$

Another simplification

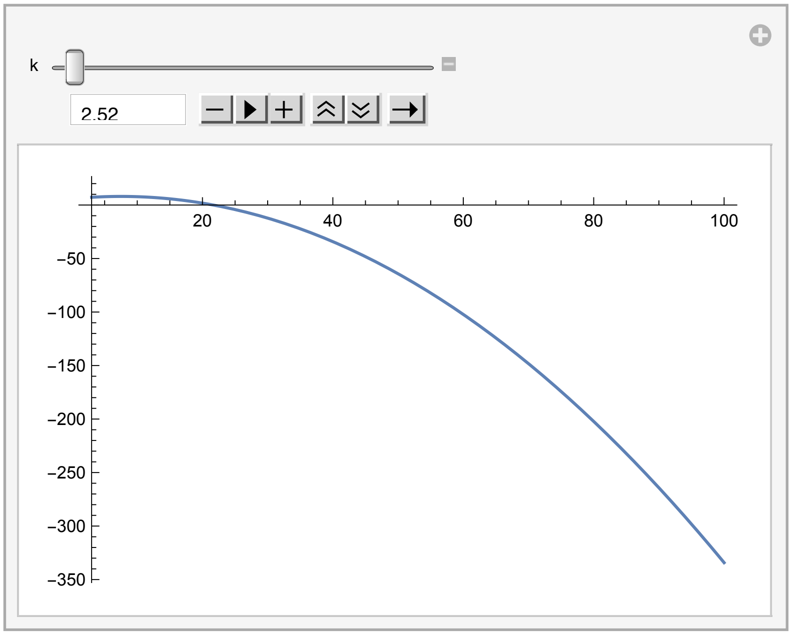
In[62]:= $$\text{FullSimplify}\left[\left(7+k\,(-1+(-3+k)\,k)-10\,n+4\,k\,n-(-3+k)\,n^2\right)>n\,(-2+k)\,(-1+n)\right]$$

Out[62]= $$7+k\,(-1+(-3+k)\,k)+5\,k\,n+(5-2\,k)\,n^2>12\,n$$

In[63]:=
```
differencea[k_, n_] := n^2 (5 - 2 k) + n (5 k - 12) + k (k - 3) + 7
```

Manipulating the parameter values of k reveals that this term is positive for k ≤ 2 and negative for k ≥3.

In[64]:= Manipulate[Plot[differencea[k, n], {n, 3, 100}], {k, 1, 100}]

Out[64]=


Understanding Boundaries of Thresholds: [b/c , 1]

$$T\_\{U\_\{ns\}\} = \frac{c\,(-1+k)\,k+b\,\left(-k\,(-2+n)+(-1+n)^2+k^2\,(-3+2\,n)\right)}{c\,\left((-1+k)\,k+(-1+n)^2\right)}.$$

For T_{U_{ns}} < 1, we found the following equivalent condition in the paper:
(n-1)^3 - k^2 (2n-3) + k(n-2) > 0
Numerically analyzing it shows that the condition does not hold for sufficiently high k relative to n:

In[65]:= cond[k_, n_] := (n - 1) ^ 3 - k ^ 2 (2 n - 3) + k (n - 2)

In[66]:= cond[6, 10]

Out[66]= 165

In[67]:= cond[7, 10]

Out[67]= - 48

The condition T_{U_{ns}(k-1)} =
$$\frac{c\,(-2+k)\,(-1+k)+b\,\left(-4+(8-3\,k)\,k+n+k\,(-5+2\,k)\,n+n^2\right)}{c\,(3+(-3+k)\,k+(-2+n)\,n)}$$
> b/c is equivalent to

In[68]:= FullSimplify[n ^ 2 + k n (2 k - 5) + n + k (8 - 3 k) - 4 > n (n - 2) + k (k - 3) + 3 - n (k - 2) (k - 1)]

Out[68]= (- 1 + k) (7 - 5 n + k (- 4 + 3 n)) > 0

In[69]:= conk[k_, n_] := 7 - 5 n + k (- 4 + 3 n)

In[70]:= conk[1, 10]

Out[70]= - 17

In[71]:= conk[2, 10]

Out[71]= 9

We found that _{U_{ns}(k-1)} < 1 is equivalent to:

In[72]:= cont[k_, n_] := n^2 (n - 3) + n (5 k + 4) - k^2 (2 n - 3) - 8 k

In[73]:= cont[7, 10]

Out[73]= 201

In[74]:= cont[8, 10]

Out[74]= -12

Proposition 4

In[75]:= FullSimplify[derUsazero[b, c, n, 3, η]]

Out[75]= $$\frac{b\,(3-n)\,(b\,(5+n)-c\,(-1+n)\,\eta)}{(b-c)\,c\,(-1+n)}$$

In[76]:= FullSimplify[derUnazero[b, c, n, 2, η]]

Out[76]= $$\frac{b\,(7\,b-2\,c-b\,n\,(4+n)+c\,(3+(-2+n)\,n)\,\eta)}{(b-c)\,c\,(-1+n)}$$

In[77]:= FullSimplify[b (3 - n) (b (5 + n) - c (-1 + n) η) > b (7 b - 2 c - b n (4 + n) + c (3 + (-2 + n) n) η)]

Out[77]= b (c + b (4 + n) - c n η) > 0

We solved for η in the paper and find that coalition size contracts at η < (2N+4)/N^2 under small alpha assumption.
We call this threshold T_2, since the coalition size contracts to 2.

T_2 > T_{U_{ns}(k-1)} if
$$\frac{2\,n+4}{n\verb|^|2} > \left(c\,(-2+k)\,(-1+k)+b\left(-4+(8-3\,k)\,k+n+k\,(-5+2\,k)\,n+n^2\right)\right)\Big/\left(c\left(3+(-3+k)\,k+(-2+n)\,n\right)\right)$$
Getting rid of the b and c as usual using bn/c = 1, we get the following:

In[78]:= FullSimplify$\left[\frac{2\,n+4}{n}\right.$ - (n (k - 2) (k - 1)) / (3 + (-3 + k) k + (-2 + n) n) > $\left(-4+(8-3\,k)\,k+n+k\,(-5+2\,k)\,n+n^2\right)$ / (3 + (-3 + k) k + (-2 + n) n)]

Out[78]= $$2+\frac{4}{n}>\frac{(-1+n)\,(4+k\,(-8+3\,k)+n)}{3+(-3+k)\,k+(-2+n)\,n}$$

In[79]:= term2[k_, n_] := 2 + $\frac{4}{n}$ - ((-1 + n) (4 + k (-8 + 3 k) + n)) / (3 + (-3 + k) k + (-2 + n) n)

Around k = 3

In[80]:= term2[3, n]

Out[80]= $$2+\frac{4}{n}-\frac{(-1+n)\,(7+n)}{3+(-2+n)\,n}$$

The term above is used for the **equation (A14)** in the paper.

T_2 > T_{U_{ns}} if

$$\frac{2n+4}{n^2} > \left(c\ (-1+k)\ k + b\ \left(-k\ (-2+n) + (-1+n)^2 + k^2\ (-3+2n)\right)\right) \Big/ \left(c\ \left((-1+k)\ k + (-1+n)^2\right)\right)$$

Getting rid of the b and c as usual using bn/c = 1, we get the following:

In[81]:= $\text{FullSimplify}\left[\frac{2n+4}{n} - \frac{n\,k\,(k-1)}{k\,(k-1)\ +\ (n-1)^{\wedge}2} > \frac{-k\,(-2+n) + (-1+n)^2 + k^2\,(-3+2n)}{(-1+k)\,k + (-1+n)^2}\right]$

Out[81]= $2 + \frac{4}{n} > \frac{(-1+n)\,(-1+k\,(-2+3k)+n)}{(-1+k)\,k + (-1+n)^2}$

In[82]:= $\text{term3}[k\_, n\_] := \text{FullSimplify}\left[2 + \frac{4}{n} - \frac{(-1+n)\,(-1+k\,(-2+3k)+n)}{(-1+k)\,k + (-1+n)^2}\right]$

Around k = 3

In[83]:= Factor[term3[3, n]]

Out[83]= $\frac{28 + 26n - 19n^2 + n^3}{n\,(7 - 2n + n^2)}$

The terms above are used in **equation (A16)** and in the previous inequality in the paper.

In[84]:= term3[3, 18]

Out[84]= $\frac{86}{2655}$

In[85]:= term3[3, 17]

Out[85]= $-\frac{54}{2227}$

T_H - T_2:

In[86]:= $\text{FullSimplify}\left[\frac{c\,(-1+k) + b\,(n + k\,(-3+k+n))}{c\,(-2+k+n)} - \frac{2n+4}{n^{\wedge}2}\right]$

Out[86]= $-\frac{2\,(2+n)}{n^2} + \frac{c\,(-1+k) + b\,(n + k\,(-3+k+n))}{c\,(-2+k+n)}$

The term above is used in **equation (A17)** in the paper. After some modification in the paper, we obtained the following:

In[87]:= `term6[k_, n_] := ((n (k - 1) + (n + k (n + k - 3))) / (n + k - 2)) > (2 (n + 2)) / n`

Around k = 3

In[88]:= FullSimplify[term6[3, n]]

Out[88]= $(-2+n)\,n\,(1+n)\,(1+2n) > 0$

T_H - T_{U_{ns}} if

In[89]:= FullSimplify$\left[\frac{c\,(-1+k)+b\,(n+k\,(-3+k+n))}{c\,(-2+k+n)}-\frac{c\,(-1+k)\,k+b\left(-k\,(-2+n)+(-1+n)^2+k^2\,(-3+2\,n)\right)}{c\left((-1+k)\,k+(-1+n)^2\right)}\right]$

Out[89]= $$\frac{(1+k-n)\,(c\,(-1+k+n-k\,n)+b\,(2-2\,n+k\,(-2+(-2+k)\,k-(-4+n)\,n)))}{c\left((-1+k)\,k+(-1+n)^2\right)(-2+k+n)}$$

Stability of Grand Coalition (Result 1)

We need to make sure that the parameters we choose satisfy:

1. b n = c
2. c > 2α
3. b > α η
4. η in [b/c,1]
5. Efforts in [0,1]

Let us redefine the effort levels and indirect utility functions by substituting b = c/n, so we integrate the first constraint into the problem:

In[90]:= qsg[c_, n_, k_, a_, η_] = FullSimplify$\left[\left((c/n)\,c\,k\,(-1+n)+(c/n)\,n\left(\frac{a\,c}{c-c/n}\right)+\left(\frac{a\,c}{c-c/n}\right)\left(\left(\frac{a\,c}{c-c/n}\right)-2\left(\frac{a\,c}{c-c/n}\right)\eta+c\,(-1+k-(-2+k+n)\,\eta)\right)\right)\Big/\left(c^2\,(-1+n)-c\,(-3+k+n)\left(\frac{a\,c}{c-c/n}\right)-2\left(\frac{a\,c}{c-c/n}\right)^2\right)\right]$

Out[90]= $$\frac{c^2\,k\,(-1+n)^3+a^2\,n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta)}{n\left(c^2\,(-1+n)^3-2\,a^2\,n^2-a\,c\,(-1+n)\,n\,(-3+k+n)\right)}$$

Defining the conditional function, not allowing efforts to take values beyond [0,1].

In[91]:= qsp[c_, n_, k_, a_, η_] :=
Piecewise$\Big[\Big\{\Big\{1,\ \left(c^2\,k\,(-1+n)^3+a^2\,n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta)\right)\Big/\left(n\left(c^2\,(-1+n)^3-2\,a^2\,n^2-a\,c\,(-1+n)\,n\,(-3+k+n)\right)\right)>1\Big\},$
$\Big\{\left(c^2\,k\,(-1+n)^3+a^2\,n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta)\right)\Big/\left(n\left(c^2\,(-1+n)^3-2\,a^2\,n^2-a\,c\,(-1+n)\,n\,(-3+k+n)\right)\right),$
$0<=\left(c^2\,k\,(-1+n)^3+a^2\,n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta)\right)\Big/\left(n\left(c^2\,(-1+n)^3-2\,a^2\,n^2-a\,c\,(-1+n)\,n\,(-3+k+n)\right)\right)\le 1\Big\},$
$\Big\{0,\ \left(c^2\,k\,(-1+n)^3+a^2\,n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta)\right)\Big/\left(n\left(c^2\,(-1+n)^3-2\,a^2\,n^2-a\,c\,(-1+n)\,n\,(-3+k+n)\right)\right)<0\Big\}\Big\}\Big]$

In[92]:= qng[c_, n_, k_, a_, η_] = FullSimplify[((c / n) (-1 + n) - (-1 + n) (a c/(c - c / n)) η + (k (a c/(c - c / n)) ((c / n) c k (-1 + n) + (c / n) n (a c/(c - c / n)) + (a c/(c - c / n)) (((a c/(c - c / n)) - 2 (a c/(c - c / n)) η + c (-1 + k - (-2 + k + n) η))))) / (c^2 (-1 + n) - c (-3 + k + n) (a c/(c - c / n)) - 2 (a c/(c - c / n))^2)) / (c (-1 + n) + (1 + k - n) (a c/(c - c / n)))]

Out[92]=
$$\frac{\frac{c(-1+n)}{n} - a\,n\,\eta + \frac{a\,k\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))}}{c\,(-1+n) + \frac{a\,(1+k-n)\,n}{-1+n}}$$

Defining the conditional function, not allowing efforts to take values beyond [0,1].

In[93]:= qnp[c_, n_, k_, a_, η_] :=

$$\text{Piecewise}\Big[\Big\{\Big\{1, \frac{\frac{c(-1+n)}{n} - a\,n\,\eta + \frac{a\,k\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))}}{c\,(-1+n) + \frac{a\,(1+k-n)\,n}{-1+n}} > 1\Big\},$$

$$\Big\{\frac{\frac{c(-1+n)}{n} - a\,n\,\eta + \frac{a\,k\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))}}{c\,(-1+n) + \frac{a\,(1+k-n)\,n}{-1+n}},$$

$$0 <= \frac{\frac{c(-1+n)}{n} - a\,n\,\eta + \frac{a\,k\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))}}{c\,(-1+n) + \frac{a\,(1+k-n)\,n}{-1+n}} \le 1\Big\},$$

$$\Big\{0, \frac{\frac{c(-1+n)}{n} - a\,n\,\eta + \frac{a\,k\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))}}{c\,(-1+n) + \frac{a\,(1+k-n)\,n}{-1+n}} < 0\Big\}\Big\}\Big]$$

In[94]:= Usg[c_, n_, k_, a_, η_] =
FullSimplify[Us[c / n, c, n, k, (a c/(c - c / n)), η, qsg[c, n, k, a, η], qng[c, n, k, a, η]]]

Out[94]=
$$-\frac{c\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))^2}{2\,n^2\,(-c^2\,(-1+n)^3 + 2\,a^2\,n^2 + a\,c\,(-1+n)\,n\,(-3+k+n))^2} +$$
$$\frac{1}{n}c\left(\frac{k\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{n\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))} + \frac{(-k+n)\left(\frac{c(-1+n)}{n} - a\,n\,\eta + \frac{a\,k\,(c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))}\right)}{c\,(-1+n) + \frac{a\,(1+k-n)\,n}{-1+n}}\right) +$$
$$\frac{1}{-1+n}a\,n\left(1 - \eta + \frac{c^2\,k\,(-1+n)^3 + a^2\,n^3\,(1-2\,\eta) - a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta)}{n\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))}\right)$$
$$\Big(-\eta + \big(c^3\,(-1+n)^4\,((-2+k)\,k+n) + a^3\,n^4\,(1-2\,(1+k-n)\,\eta) + a^2\,c\,(-1+n)\,n^2\,(-3\,n+2\,k\,(1+n)+n\,(-2-k+n)\,(-2+k+n)\,\eta) - a\,c^2\,(-1+n)^2\,n\,(k^2\,(-1+n\,(-1+\eta)) + k\,(4-2\,n\,\eta) + n\,(-3+n+(2+(-2+n)\,n)\,\eta))\big) \big/ \big(n\,(c\,(-1+n)^2 + a\,(1+k-n)\,n)\,(c^2\,(-1+n)^3 - 2\,a^2\,n^2 - a\,c\,(-1+n)\,n\,(-3+k+n))\big)\Big)$$

In[95]:=

```
Ung[c_, n_, k_, a_, η_] =
  FullSimplify[Un[c / n, c, n, k, (a c)/(c - c / n), η, qsg[c, n, k, a, η], qng[c, n, k, a, η]]]
```

Out[95]=

$$-\frac{c\left(\frac{c(-1+n)}{n}-a\,n\,\eta+\frac{a\,k\,(c^2 k\,(-1+n)^3+a^2 n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3-2\,a^2 n^2-a\,c\,(-1+n)\,n\,(-3+k+n))}\right)^2}{2\left(c\,(-1+n)+\frac{a\,(1+k-n)\,n}{-1+n}\right)^2}+$$

$$\frac{1}{-1+n}a\,n\Big(-\eta+\Big(c^3\,(-1+n)^4\,(-1+(-1+k)\,k+n)+2\,a^3 n^4\,(-1-k+n)\,\eta-$$
$$a\,c^2\,(-1+n)^2\,n\,\big(3+k^2\,(-1+n\,(-1+\eta))+n\,\big(-4+n+(-1+n)^2\,\eta\big)+k\,(2-n\,\eta)\big)+$$
$$a^2 c\,(-1+n)\,n^2\,\big(k\,(2+n)-k^2 n\,\eta+(-1+n)\,(-2+(-3+n)\,n\,\eta)\big)\Big)\Big/$$
$$\Big(n\,\big(c\,(-1+n)^2+a\,(1+k-n)\,n\big)\,\big(c^2\,(-1+n)^3-2\,a^2 n^2-a\,c\,(-1+n)\,n\,(-3+k+n)\big)\Big)\Big)$$

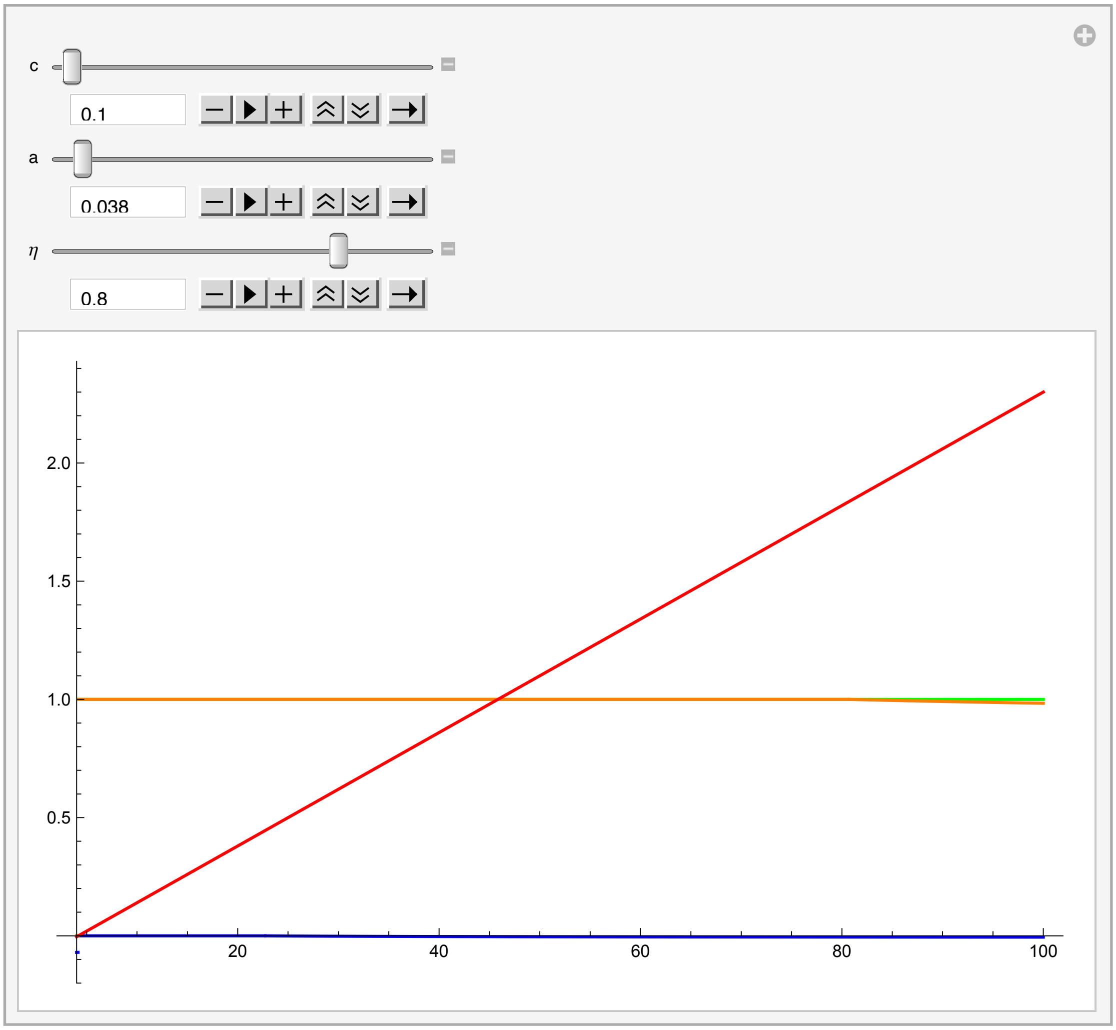
$$\left(1-\eta+\frac{\frac{c(-1+n)}{n}-a\,n\,\eta+\frac{a\,k\,(c^2 k\,(-1+n)^3+a^2 n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3-2\,a^2 n^2-a\,c\,(-1+n)\,n\,(-3+k+n))}}{c\,(-1+n)+\frac{a\,(1+k-n)\,n}{-1+n}}\right)+\frac{1}{n}$$

$$c\left(\frac{k\left(c^2 k\,(-1+n)^3+a^2 n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta)\right)}{n\left(c^2\,(-1+n)^3-2\,a^2 n^2-a\,c\,(-1+n)\,n\,(-3+k+n)\right)}+\right.$$
$$\left.\frac{(-k+n)\left(\frac{c(-1+n)}{n}-a\,n\,\eta+\frac{a\,k\,(c^2 k\,(-1+n)^3+a^2 n^3\,(1-2\,\eta)-a\,c\,(-1+n)\,n^2\,(-k+(-2+k+n)\,\eta))}{(-1+n)\,(c^2\,(-1+n)^3-2\,a^2 n^2-a\,c\,(-1+n)\,n\,(-3+k+n))}\right)}{c\,(-1+n)+\frac{a\,(1+k-n)\,n}{-1+n}}\right)$$

Analyzing the condition U_s(n) ≥ U_n(n-1), restricting the efforts level to be in [0,1] using the piecewise functions:

```
In[96]:= Manipulate[Plot[{Us[c / n, c, n, n, a c/(c - c / n), η, qsp[c, n, n, a, η], qnp[c, n, n, a, η]] -
      Un[c / n, c, n, n - 1, a c/(c - c / n), η, qsp[c, n, n - 1, a, 0.8], qnp[c, n, n - 1, a, η]],
     qsp[c, n, n, a, η], qnp[c, n, n, a, η], c (n - 1) - 2 a n}, {n, 4, 100},
   PlotStyle → {Blue, Green, Orange, Red}], {c, 0.01, 10}, {a, 0, 1}, {η, 0.1, 1}]
```

Out[96]=



While the condition U_s(n) ≥ U_n(n-1) hold with equality on the Figure above, let us make sure that it is indeed the case.

```
In[97]:= Us[0.1 / 100, 0.1, 100, 100, 0.038 × 0.1/(0.1 - 0.1 / 100), 0.799, 1, 1] -
     Un[0.1 / 100, 0.1, 100, 100, 0.038 × 0.1/(0.1 - 0.1 / 100), .799, 1, 1]

Out[97]= 0.
```

In[98]:=
```
Us[1 / 100, 1, 100, 100, (0.436 × 1)/(1 - 1 / 100), 0.799, 1, 1] -
  Un[1 / 100, 1, 100, 100, (0.436 × 1)/(1 - 1 / 100), 0.799, 1, 1]
```
Out[98]= 0.

In[99]:= `qsp[0.1, 100, 100, 0.038, 0.8]`

Out[99]= 1

In[100]:= `qnp[.1, 100, 100, 0.038, 0.8]`

Out[100]= 0.983634

In[101]:= `qsp[0.1, 100, 99, 0.038, 0.8]`

Out[101]= 1

In[102]:= `qnp[.1, 100, 99, 0.038, 0.8]`

Out[102]= 0.936364

Strategic Use of Eta

In this section, we assume N-1 countries set their truthful expectations “eta” while one country is strategically setting its expectations. Thus, we have a case of heterogenous expectations. In this supporting file, we first solve the game with possibly two types of countries different in their expectations, which we call them low and high expectations here. Then, assume that there is only one country of one of the types.

Let the number of countries be N, low types be l and, thus, the high types be N-l. Among k signatories, k_L and k_H be the number of low and high types, respectively.
The utility under symmetry, the equation (7): U(b, c, α, η, N, q_i, Q_-i) = b(Q_-i + qi) - (c/2)(qi)^2 +α(Q_-i/(N -1) - η)(q_i +1 - η)
Let’s solve the non-cooperative game.

The utility function of a country with LOW EXPECTATIONS is:

In[103]:=
```
utilitynoncooplowetai[b_, c_, α_, ηL_, ηH_, n_, l_, qLn_, qHn_, qLi_] :=
  b ((n - l) qHn + (l - 1) qLn + qLi) - (c / 2) qLi^2 +
   α (((n - l) qHn + (l - 1) qLn) / (n - 1) - ηL) (qLi + 1 - ηH)
```

The best-response function of country i with LOW EXPECTATION:

In[104]:=
```
FullSimplify[
  Solve[D[utilitynoncooplowetai[b, c, α, ηL, ηH, n, l, qLn, qHn, qLi], qLi] == 0, qLi]]
```

Out[104]= $\left\{\left\{qLi \to \frac{b + \frac{\alpha\,(n\,qHn - qLn + l\,(-qHn + qLn) + \eta L - n\,\eta L)}{-1+n}}{c}\right\}\right\}$

In equilibrium, all identical types exert the same effort level:

In[105]:=
```
FullSimplify[
 Solve[qLn == (b + (α (n qHn - qLn + l (-qHn + qLn) + ηL - n ηL)) / (-1 + n)) / c, qLn]]
```
Out[105]=
$$\left\{\left\{qLn \to \frac{b\,(-1+n)+\alpha\,(-l\,qHn+n\,qHn+\eta L-n\,\eta L)}{c\,(-1+n)+\alpha-l\,\alpha}\right\}\right\}$$

The utility function of a country with HIGH EXPECTATIONS is:

In[106]:=
```
utilitynoncoophighetai[b_, c_, α_, ηL_, ηH_, n_, l_, qLn_, qHn_, qHi_] :=
 b ((n - l - 1) qHn + l qLn + qHi) - (c / 2) qHi^2 +
  α (((n - l - 1) qHn + l qLn) / (n - 1) - ηH) (qHi + 1 - ((n - 2) / (n - 1)) ηH - ηL / (n - 1))
```

The best-response function of country i with HIGH EXPECTATIONS:

In[107]:=
```
FullSimplify[Solve[
  D[utilitynoncoophighetai[b, c, α, ηL, ηH, n, l, qLn, qHn, qHi], qHi] == 0, qHi]]
```
Out[107]=
$$\left\{\left\{qHi \to \frac{b+\frac{\alpha\,((-1-l+n)\,qHn+l\,qLn+\eta H-n\,\eta H)}{-1+n}}{c}\right\}\right\}$$

In equilibrium, all identical types exert the same effort level:

In[108]:=
```
FullSimplify[
 Solve[qHn == (b + (α ((-1 - l + n) qHn + l qLn + ηH - n ηH)) / (-1 + n)) / c, qHn]]
```
Out[108]=
$$\left\{\left\{qHn \to \frac{b\,(-1+n)+\alpha\,(l\,qLn+\eta H-n\,\eta H)}{c\,(-1+n)+(1+l-n)\,\alpha}\right\}\right\}$$

Substituting one into the other yields the non-cooperative efforts with HIGH and LOW EXPECTATIONS:

In[109]:=
$$\text{FullSimplify}\left[\text{Solve}\left[qHn == \frac{b\,(-1+n)+\alpha\left(l\left(\frac{b\,(-1+n)+\alpha\,(-l\,qHn+n\,qHn+\eta L-n\,\eta L)}{c\,(-1+n)+\alpha-l\,\alpha}\right)+\eta H-n\,\eta H\right)}{c\,(-1+n)+(1+l-n)\,\alpha},\ qHn\right]\right]$$
Out[109]=
$$\left\{\left\{qHn \to \frac{b\,(c\,(-1+n)+\alpha)+\alpha\,((c-c\,n+(-1+l)\,\alpha)\,\eta H-l\,\alpha\,\eta L)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}\right\}\right\}$$

In[110]:=
$$\text{FullSimplify}\left[\text{Solve}\left[qLn == \frac{b\,(-1+n)+\alpha\left((n-l)\left(\frac{b\,(-1+n)+\alpha\,(l\,qLn+\eta H-n\,\eta H)}{c\,(-1+n)+(1+l-n)\,\alpha}\right)+\eta L-n\,\eta L\right)}{c\,(-1+n)+\alpha-l\,\alpha},\ qLn\right]\right]$$
Out[110]=
$$\left\{\left\{qLn \to \frac{b\,(c\,(-1+n)+\alpha)+\alpha\,(-n\,\alpha\,\eta H+l\,\alpha\,(\eta H-\eta L)+(-1+n)\,(-c+\alpha)\,\eta L)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}\right\}\right\}$$

In[111]:=
$$\text{noncoopeffhigh}[c\_,\ \alpha\_,\ \eta H\_,\ \eta L\_,\ n\_,\ l\_] := \frac{(c/n)\,(c\,(-1+n)+\alpha)+\alpha\,((c-c\,n+(-1+l)\,\alpha)\,\eta H-l\,\alpha\,\eta L)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}$$

In[112]:=
```
noncoopefflow[c_, α_, ηH_, ηL_, n_, l_] :=
 ((c / n) (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL)) /
  ((c - α) (c (-1 + n) + α))
```

Let the number of “low” types be 1, then the effort levels become as follows (**Equations 20 and 21 in the paper**):

In[113]:=
```
noncoopeffhigh[c, α, ηH, ηL, n, 1]
```
Out[113]=
$$\frac{\frac{c\,(c\,(-1+n)+\alpha)}{n}+\alpha\,((c-c\,n)\,\eta H-\alpha\,\eta L)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}$$

In[114]:= noncoopefflow[c, α, ηH, ηL, n, 1]

Out[114]=
$$\frac{\frac{c\,(c\,(-1+n)+\alpha)}{n}+\alpha\,(-n\,\alpha\,\eta H+\alpha\,(\eta H-\eta L)+(-1+n)\,(-c+\alpha)\,\eta L)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}$$

From the best-responses qLn and qHn, one can see that qLn > qHn since ηH > ηL. Let us give some parameter values to understand how etas affect effort levels.

In[115]:= FullSimplify[noncoopeffhigh[5, 0.1, ηH, ηL, 10, 1]]

Out[115]= $0.102041 - 0.0203629\,\eta H - 0.0000452509\,\eta L$

In[116]:= FullSimplify[noncoopefflow[5, 0.1, ηH, ηL, 10, 1]]

Out[116]= $0.102041 - 0.000407258\,\eta H - 0.0200009\,\eta L$

Self-fulfilling Prophecy

Let us find the indirect utility levels. First, a country strategically sets its expectation under self-fulfilling prophecy assumption:

```
In[117]:= indutilnoncooploweta[c_, α_, ηH_, ηL_, n_, l_] :=
  FullSimplify[(c / n) ((n - l) (((c / n) (c (-1 + n) + α) + α ((c - c n + (-1 + l) α) ηH - l α ηL)) /
            ((c - α) (c (-1 + n) + α))) + l (((c / n) (c (-1 + n) + α) +
              α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL)) / ((c - α) (c (-1 + n) + α)))) -
     (c / 2) (((c / n) (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL)) /
          ((c - α) (c (-1 + n) + α))) ^2 +
     α (((n - l) (((c / n) (c (-1 + n) + α) + α ((c - c n + (-1 + l) α) ηH - l α ηL)) /
                ((c - α) (c (-1 + n) + α))) +
            (l - 1) (((c / n) (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL)) /
                ((c - α) (c (-1 + n) + α)))) / (n - 1) - ηL)
      ((((c / n) (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL)) /
          ((c - α) (c (-1 + n) + α))) + 1 - ηH)]
```

Country S' optimal strategic eta:

In[118]:= FullSimplify[Solve[D[indutilnoncooploweta[c, α, ηH, ηL, n, 1], ηL] == 0, ηL]]

Out[118]=
$$\Big\{\Big\{\eta L\to\Big(-c^3\,(-1+n)^2\,(-1+n\,(-1+\eta H))+2\,c^2\,(-1+n)\,\alpha\,(1+(-2+n)\,n\,(-1+\eta H))+\alpha^3\,\big(1+n\,\big(-3+n+(-2+n)^2\,\eta H\big)\big)+c\,n\,\alpha^2\,(7-6\,\eta H+n\,(-6+n+7\,\eta H-2\,n\,\eta H))\Big)\Big/\Big(n\,\alpha\,(c-c\,n+(-2+n)\,\alpha)^2\Big)\Big\}\Big\}$$

In[119]:=
$$\text{streta}[c\_,\alpha\_,\eta H\_,n\_]:=\frac{1}{n\,\alpha\,(c-c\,n+(-2+n)\,\alpha)^2}\Big(-c^3\,(-1+n)^2\,(-1+n\,(-1+\eta H))+2\,c^2\,(-1+n)\,\alpha\,(1+(-2+n)\,n\,(-1+\eta H))+\alpha^3\,\big(1+n\,\big(-3+n+(-2+n)^2\,\eta H\big)\big)+c\,n\,\alpha^2\,(7-6\,\eta H+n\,(-6+n+7\,\eta H-2\,n\,\eta H))\Big)$$

In[120]:= Collect[streta[c, α, ηH, n], ηH]

Out[120]=
$$\Big(c^3\,(-1+n)^2+c^3\,(-1+n)^2\,n+2\,c^2\,(-1+n)\,\alpha-2\,c^2\,(-2+n)\,(-1+n)\,n\,\alpha+7\,c\,n\,\alpha^2-6\,c\,n^2\,\alpha^2+c\,n^3\,\alpha^2+\alpha^3-3\,n\,\alpha^3+n^2\,\alpha^3\Big)\Big/\Big(n\,\alpha\,(c-c\,n+(-2+n)\,\alpha)^2\Big)+\frac{\Big(-c^3\,(-1+n)^2\,n+2\,c^2\,(-2+n)\,(-1+n)\,n\,\alpha-6\,c\,n\,\alpha^2+7\,c\,n^2\,\alpha^2-2\,c\,n^3\,\alpha^2+(-2+n)^2\,n\,\alpha^3\Big)\,\eta H}{n\,\alpha\,(c-c\,n+(-2+n)\,\alpha)^2}$$

The first two equations below show the optimal strategic eta "ηL" for some specific parameter values and in particular how it changes with the truthful expectations "ηH". The last two equations show the effort levels of the strategic and non-strategic countries. We evaluate these equations for a wide range of parameters and shared our findings on **equation (18), result 2 and the paragraph between the two.**

In[121]:= `FullSimplify[streta[.2, 0.09, ηH, 20]]`

Out[121]= $2.46474 - 2.96534\,\eta H$

In[122]:= `FullSimplify[streta[.2, 0.1, 0.9, 10]]`

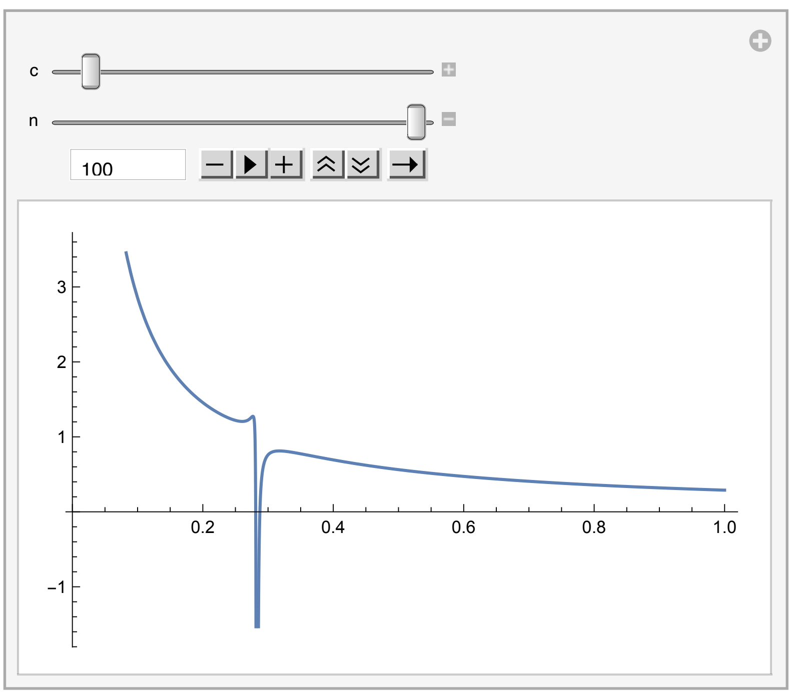
Out[122]= -0.069

Let us observe the exact terms for intercept and the slope to better understand each:

In[123]:= `Collect[streta[c, α, ηH, n], ηH]`

Out[123]=
$$\left(c^3 (-1+n)^2 + c^3 (-1+n)^2 n + 2 c^2 (-1+n)\alpha - 2 c^2 (-2+n)(-1+n) n \alpha + 7 c n \alpha^2 - 6 c n^2 \alpha^2 + c n^3 \alpha^2 + \alpha^3 - 3 n \alpha^3 + n^2 \alpha^3\right) \Big/ \left(n \alpha \left(c - c n + (-2+n)\alpha\right)^2\right) + \frac{\left(-c^3 (-1+n)^2 n + 2 c^2 (-2+n)(-1+n) n \alpha - 6 c n \alpha^2 + 7 c n^2 \alpha^2 - 2 c n^3 \alpha^2 + (-2+n)^2 n \alpha^3\right) \eta H}{n \alpha \left(c - c n + (-2+n)\alpha\right)^2}$$

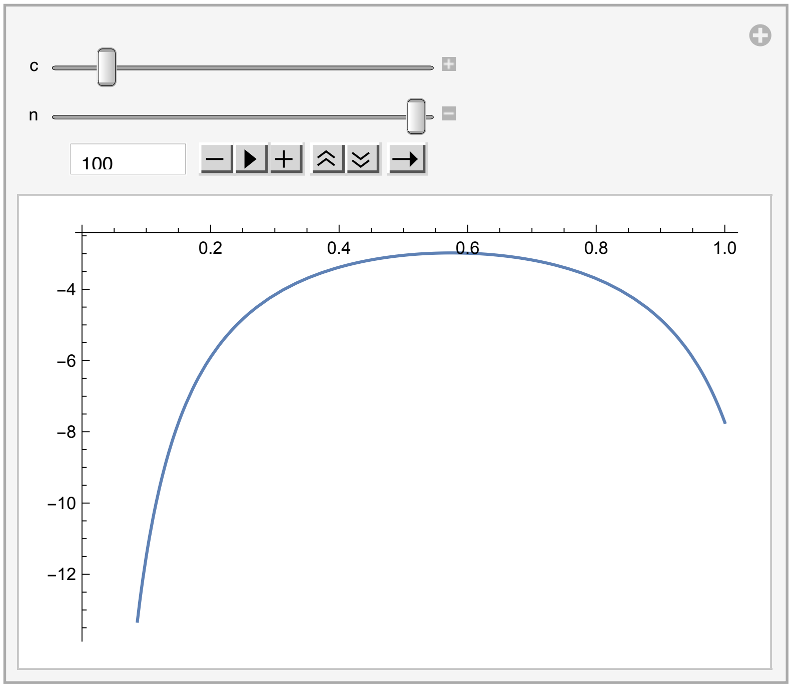
In[124]:= `Manipulate[Plot[(c^3 (-1 + n)^2 + c^3 (-1 + n)^2 n + 2 c^2 (-1 + n) α - 2 c^2 (-2 + n) (-1 + n) n α + 7 c n α^2 - 6 c n^2 α^2 + c n^3 α^2 + α^3 - 3 n α^3 + n^2 α^3) / (n α (c - c n + (-2 + n) α)^2), {α, 0.01, 1}], {c, 0.1, 3}, {n, 5, 100}]`

Out[124]=


Keeping in mind that we assume c > 2α, the intercept term of the strategic eta, ηL, always weakly decreasing in alpha.
Increase in n or lower b also reduces the intercept term.

In[125]:=
```
Manipulate[Plot[
  (-c^3 (-1 + n)^2 n + 2 c^2 (-2 + n) (-1 + n) n α - 6 c n α^2 + 7 c n^2 α^2 - 2 c n^3 α^2 + (-2 + n)^2 n α^3) /
   (n α (c - c n + (-2 + n) α)^2), {α, 0.01, 1}], {c, 0.1, 10}, {n, 5, 100}]
```

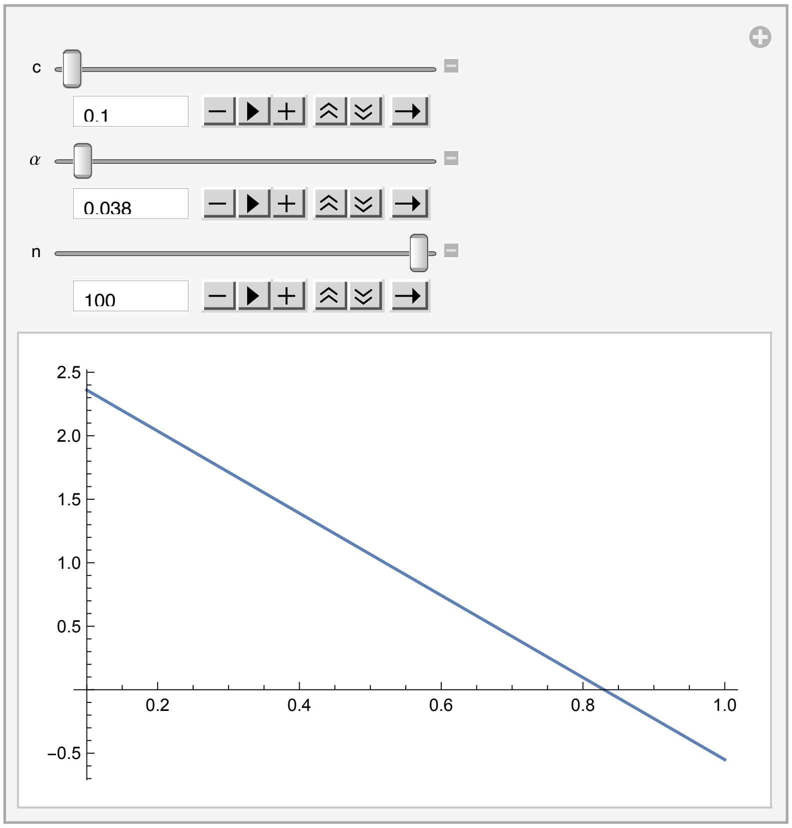
Out[125]=



Keeping in mind that we assume c > 2α, the magnitude of the slope of the strategic eta, ηL, always weakly decreasing in alpha.

```
In[126]:= Manipulate[Plot[streta[c, α, ηH, n], {ηH, 0.1, 1}],
  {c, 0.1, 3}, {α, 0.01, 1}, {n, 5, 100}]
```

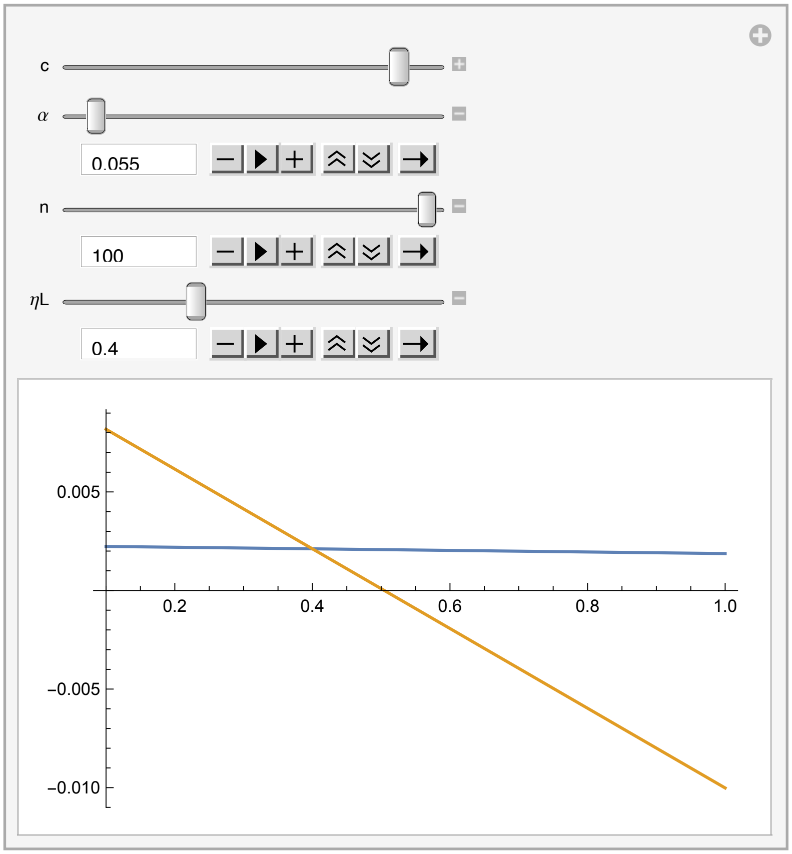
Out[126]=




```
In[127]:= FullSimplify[noncoopeffhigh[1, 0.28, ηH, ηL, 10, 1]]

Out[127]= 0.138889 - 0.377155 ηH - 0.0117337 ηL

In[128]:= FullSimplify[noncoopefflow[8, 0.1, ηH, ηL, 10, 1]]

Out[128]= 0.101266 - 0.000158008 ηH - 0.0125002 ηL

In[129]:= FullSimplify[noncoopeffhigh[0.6, 0.28, 0.6, ηL, 100, 1]]

Out[129]= -0.503787 - 0.00410523 ηL

In[130]:= FullSimplify[noncoopefflow[0.6, 0.28, 0.6, ηL, 10, 1]]

Out[130]= -0.0454225 - 0.486796 ηL
```

In[131]:=
```
Manipulate[
 Plot[{noncoopefflow[c, α, ηH, ηL, n, 1], noncoopeffhigh[c, α, ηH, ηL, n, 1]},
  {ηH, 0.1, 1}], {c, 0.1, 3}, {α, 0.01, 1}, {n, 5, 100}, {ηL, 0.1, 1}]
```

Out[131]=



Heterogeneity: Self-Interested and Reciprocal Countries

There are A<N reciprocal countries with α_i = α and N-A self-interested countries with α_i = 0. With abuse of notation, we avoid using the capital letter “A” and instead use the lower case letter “a”. This “a” is different than the one that capture how much a country care about reciprocity in comparison to material pay-offs. In this section, we only use α

Non-Cooperative Game - Proposition 5

Self-interested countries have a dominant strategy to exert b/c. Let’s find out about the reciprocal Nash efforts. To do that, let’s write the utility function.

In[132]:= ui[b_, c_, n_, a_, α_, η_, qi_, qj_] :=

$$b\left((n-a)\frac{b}{c}+(a-1)\,qj+qi\right)-\frac{c}{2}\,qi^2+\alpha\left(\frac{(n-a)\,\frac{b}{c}+(a-1)\,qj}{n-1}-\eta\right)\left(qi+1-\eta\right)$$

Taking derivative wrt qi and solving for qi yields the Nash efforts of the reciprocal countries:

In[133]:= D[ui[b, c, n, a, α, η, qi, qj], qi]

Out[133]= $$b - c\,qi + \alpha\left(\frac{\frac{b\,(-a+n)}{c} + (-1+a)\,qj}{-1+n} - \eta\right)$$

In[134]:= $$\text{FullSimplify}\left[\text{Solve}\left[b - c\,qi + \alpha\left(\frac{\frac{b\,(-a+n)}{c} + (-1+a)\,qi}{-1+n} - \eta\right) == 0,\ qi\right]\right]$$

Out[134]= $$\left\{\left\{qi \to \frac{b\,c\,(-1+n) + b\,(-a+n)\,\alpha - c\,(-1+n)\,\alpha\,\eta}{c\,(c\,(-1+n) + \alpha - a\,\alpha)}\right\}\right\}$$

In[135]:= $$\text{qnc}[b\_, c\_, n\_, a\_, \alpha\_, \eta\_] := \frac{b\,c\,(-1+n) + b\,(-a+n)\,\alpha - c\,(-1+n)\,\alpha\,\eta}{c\,(c\,(-1+n) + \alpha - a\,\alpha)}$$

What is the impact of α on qnc? Similar to the homogenous case, whether η > b/c holds determines the sign of the impact.

In[136]:= FullSimplify[D[qnc[b, c, n, a, α, η], α]]

Out[136]= $$\frac{(-1+n)^2\,(b - c\,\eta)}{(c\,(-1+n) + \alpha - a\,\alpha)^2}$$

In[137]:= $$\frac{(-1+n)^2\,(b - c\,\eta)}{(c\,(-1+n) + \alpha - a\,\alpha)^2}$$

Out[137]= $$\frac{(-1+n)^2\,(b - c\,\eta)}{(c\,(-1+n) + \alpha - a\,\alpha)^2}$$

The level of reciprocal concern α needed for the reciprocal countries to exert full effort:

In[138]:= FullSimplify[Solve[qnc[c / n, c, n, a, α, η] == 1, α]]

Out[138]= $$\left\{\left\{\alpha \to \frac{c\,(-1+n)}{a - n\,\eta}\right\}\right\}$$

The level of reciprocal concern α needed for the reciprocal countries to exert zero effort:

In[139]:= FullSimplify[Solve[qnc[c / n, c, n, a, α, η] == 0, α]]

Out[139]= $$\left\{\left\{\alpha \to \frac{c\,(-1+n)}{a + n\,(-1 + (-1+n)\,\eta)}\right\}\right\}$$

Partial Cooperation - Result 4

Reciprocal Non-signatories

Remember that self-interested non-signatories have dominant strategy of exerting b/c.
kr and ks refer to the number of reciprocal and self-interested signatories.

In[140]:=
```
urnsi[b_, c_, n_, a_, ks_, kr_, α_, η_, qi_, qss_, qrs_, qrns_] :=
```
$$b\left((n-a-ks)\,\frac{b}{c}+ks*qss+kr*qrs+(a-kr-1)\,qrns+qi\right)-\frac{c}{2}\,qi^2+\alpha\left(\frac{(n-a-ks)\,\frac{b}{c}+ks*qss+kr*qrs+(a-kr-1)\,qrns}{n-1}-\eta\right)(qi+1-\eta)$$

Taking derivative wrt qi yields:

In[141]:=
```
D[urnsi[b, c, n, a, ks, kr, α, η, qi, qss, qrs, qrns], qi]
```

Out[141]=
$$b-c\,qi+\alpha\left(\frac{\frac{b\,(-a-ks+n)}{c}+(-1+a-kr)\,qrns+kr\,qrs+ks\,qss}{-1+n}-\eta\right)$$

Using symmetry and solving for qrns yields the result:

In[142]:=
$$\text{FullSimplify}\left[\text{Solve}\left[b-c\,qrns+\alpha\left(\frac{\frac{b\,(-a-ks+n)}{c}+(a-kr-1)\,qrns+kr\,qrs+ks\,qss}{-1+n}-\eta\right)==qrns,\ qrns\right]\right]$$

Out[142]=
$$\left\{\left\{qrns\to\frac{b\,c\,(-1+n)-b\,(a+ks-n)\,\alpha+c\,\alpha\,(kr\,qrs+ks\,qss+\eta-n\,\eta)}{c\,(-1+c\,(-1+n)+n+\alpha-a\,\alpha+kr\,\alpha)}\right\}\right\}$$

In[143]:=
```
qrns[b_, c_, n_, a_, ks_, kr_, α_, η_, qss_, qrs_] :=
```
$$\frac{b\,c\,(-1+n)-b\,(a+ks-n)\,\alpha+c\,\alpha\,(kr\,qrs+ks\,qss+\eta-n\,\eta)}{c\,(-1+c\,(-1+n)+n+\alpha-a\,\alpha+kr\,\alpha)}$$

Signatories

Let's start with the self-interested signatories problem.

In[144]:=
```
usig[b_, c_, n_, a_, ks_, kr_, α_, η_, qss_, qrs_] :=
```
$$\text{FullSimplify}\Big[ks\left(b\left((n-a-ks)\,\frac{b}{c}+(a-kr)\,qrns[b,c,n,a,ks,kr,\alpha,\eta,qss,qrs]+ks*qss+kr*qrs\right)-\frac{c}{2}\,qss^2\right)+$$
$$kr\left(b\left((n-a-ks)\,\frac{b}{c}+(a-kr)\,qrns[b,c,n,a,ks,kr,\alpha,\eta,qss,qrs]+ks*qss+kr*qrs\right)-\frac{c}{2}\,qrs^2-\right.$$
$$\left.+\alpha\left(\frac{1}{n-1}\left((n-a-ks)\,\frac{b}{c}+(a-kr)\,qrns[b,c,n,a,ks,kr,\alpha,\eta,qss,qrs]+ks*qss+(kr-1)\,qrs\right)-\eta\right)(qrs+1-\eta)\right)\Big]$$

Take first derivative wrt qss (FOC) and solving for qss (self-interested signatory effort); and then similarly with qrs (reciprocal signatory effort).

In[145]:=
```
FullSimplify[Solve[D[usig[b, c, n, a, ks, kr, α, η, qss, qrs], qss] == 0, qss]]
```

Out[145]=
$$\left\{\left\{qss\to\frac{((1+c)\,(-1+n)+\alpha)\,(b\,(kr+ks)\,(-1+n)+kr\,\alpha\,(-1-qrs+\eta))}{c\,(-1+n)\,(-1+c\,(-1+n)+n+\alpha-a\,\alpha+kr\,\alpha)}\right\}\right\}$$

```
In[146]:= FullSimplify[Solve[D[usig[b, c, n, a, ks, kr, α, η, qss, qrs], qrs] == 0, qrs]]

Out[146]= {{qrs →
     (b (c (1 + c) (kr + ks) (-1 + n)^2 + (-1 + n) (a + ks + 2 c (kr + ks) - n - c n) α + (a + ks - n) α^2) +
        c α (-1 + n + ks qss + α - a α - ks qss (n + α) + kr (-1 + n) (-1 + η) + 2 η +
           (-3 + n) n η + (-2 + a + n) α η + c (-1 + n) (1 - kr - ks qss + (-2 + kr + n) η))) /
      (c (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2))}}
```

Substituting qrs into qss and solving for qss yields

```
In[147]:= FullSimplify[Solve[
     (((1 + c) (-1 + n) + α) (b (kr + ks) (-1 + n) + kr α (-1 - ((b (c (1 + c) (kr + ks) (-1 + n)^2 +
                    (-1 + n) (a + ks + 2 c (kr + ks) - n - c n) α + (a + ks - n) α^2) + c α
                 (-1 + n + ks qss + α - a α - ks qss (n + α) + kr (-1 + n) (-1 + η) + 2 η + (-3 + n)
                    n η + (-2 + a + n) α η + c (-1 + n) (1 - kr - ks qss + (-2 + kr + n) η))) /
               (c (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α +
                    2 (-1 + a) α^2))) + η))) /
       (c (-1 + n) (-1 + c (-1 + n) + n + α - a α + kr α)) == qss, qss]]

Out[147]= {{qss →
     ((1 - n) ((1 + c) (-1 + n) + α) (b (c^3 (kr + ks) (-1 + n)^3 - kr (a + ks - n) α^2 (-1 + n + α) + c^2
                (kr + ks) (-1 + n)^2 (-1 + n - (1 + a - 2 kr) α) + c (-1 + n) α
                ((-2 + kr) (kr + ks) (-1 + n) + 2 (-1 + a - kr) (kr + ks) α + kr n α)) + c kr α
            (c^2 (-1 + n)^2 (-1 + η) - c (-1 + n) (-1 + n - a α + 2 kr α + η - n η + (-1 + a - 2 kr + n)
                  α η) + α (-1 + n + α - a α + kr (-1 + n) (-1 + η) + a α η - n (-1 + n + α) η)))) /
      (c (-1 + n) (-kr ks α^2 ((1 + c) (-1 + n) + α) + c kr ks (-1 + n) α^2 ((1 + c) (-1 + n) + α) +
           kr ks α^2 ((1 + c) (-1 + n) + α) (n + α) - c (1 - n) (-(1 + c) (-1 + n) + (-1 + a - kr) α)
            (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2)))}}
```

Self-interested signatory effort

```
In[148]:= qss[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
   ((1 - n) ((1 + c) (-1 + n) + α) (b (c^3 (kr + ks) (-1 + n)^3 - kr (a + ks - n) α^2 (-1 + n + α) +
             c^2 (kr + ks) (-1 + n)^2 (-1 + n - (1 + a - 2 kr) α) + c (-1 + n) α
              ((-2 + kr) (kr + ks) (-1 + n) + 2 (-1 + a - kr) (kr + ks) α + kr n α)) + c kr α
          (c^2 (-1 + n)^2 (-1 + η) - c (-1 + n) (-1 + n - a α + 2 kr α + η - n η + (-1 + a - 2 kr + n) α η) +
             α (-1 + n + α - a α + kr (-1 + n) (-1 + η) + a α η - n (-1 + n + α) η)))) /
    (c (-1 + n) (-kr ks α^2 ((1 + c) (-1 + n) + α) + c kr ks (-1 + n) α^2 ((1 + c) (-1 + n) + α) +
         kr ks α^2 ((1 + c) (-1 + n) + α) (n + α) - c (1 - n) (-(1 + c) (-1 + n) + (-1 + a - kr) α)
          (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2)))
```

Self-interested signatory effort without b:

```
In[149]:= qssnob[c_, n_, a_, ks_, kr_, α_, η_] :=
   ((1 - n) ((1 + c) (-1 + n) + α) (c / n (c^3 (kr + ks) (-1 + n)^3 - kr (a + ks - n) α^2 (-1 + n + α) +
             c^2 (kr + ks) (-1 + n)^2 (-1 + n - (1 + a - 2 kr) α) + c (-1 + n) α
              ((-2 + kr) (kr + ks) (-1 + n) + 2 (-1 + a - kr) (kr + ks) α + kr n α)) + c kr α
          (c^2 (-1 + n)^2 (-1 + η) - c (-1 + n) (-1 + n - a α + 2 kr α + η - n η + (-1 + a - 2 kr + n) α η) +
             α (-1 + n + α - a α + kr (-1 + n) (-1 + η) + a α η - n (-1 + n + α) η)))) /
    (c (-1 + n) (-kr ks α^2 ((1 + c) (-1 + n) + α) + c kr ks (-1 + n) α^2 ((1 + c) (-1 + n) + α) +
         kr ks α^2 ((1 + c) (-1 + n) + α) (n + α) - c (1 - n) (-(1 + c) (-1 + n) + (-1 + a - kr) α)
          (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2)))
```

Substitute qss into qrs

In[150]:=
```
FullSimplify[
 (b (c (1 + c) (kr + ks) (-1 + n)^2 + (-1 + n) (a + ks + 2 c (kr + ks) - n - c n) α + (a + ks - n) α^2) +
    c α (-1 + n + ks (qss[b, c, n, a, ks, kr, α, η]) + α - a α - ks (qss[b, c, n, a, ks, kr,
          α, η]) (n + α) + kr (-1 + n) (-1 + η) + 2 η + (-3 + n) n η + (-2 + a + n) α η +
       c (-1 + n) (1 - kr - ks (qss[b, c, n, a, ks, kr, α, η]) + (-2 + kr + n) η))) /
  (c (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2))]
```

Out[150]=
```
(b (-1 + n) (c (1 + c)^2 (kr + ks) (-1 + n)^3 - (1 + c) (-1 + n)^2
         (-ks + (kr + ks) (ks + c (-3 - kr + ks)) + a (-1 + c (kr + ks)) + n + c n) α -
        (-1 + n) (a^2 + ks (-2 + kr + 2 ks) + a (-2 - kr + ks + c (-1 + 2 kr + 2 ks - n) - n) +
           (2 + kr) n + c (ks (-3 + 2 ks) + 2 n + kr (-2 - 2 kr + n))) α^2 -
        (a^2 - ks + ks^2 + n + kr n - a (1 + kr - ks + n)) α^3) +
    α (-kr ks α (-1 + n + α)^2 (-1 + η) + c^3 (-1 + n)^3 (1 - kr + (-2 + kr + n) η) +
       c^2 (-1 + n)^2 (-2 + 2 kr + 2 n - 2 kr n + 2 α - 2 a α + a kr α - kr^2 α + kr ks α +
          2 (-1 + n) (-2 + kr + n) η - (4 + kr - 2 n + a (-3 + kr + n) - kr (kr - ks + n)) α η) +
       c (-1 + n) (kr^2 (-1 + n) α (-1 + η) + (-1 + n + α - a α)
            (-1 + n + α - a α + (2 + (-2 + a) α + n (-3 + n + α)) η) +
          kr (-1 + η + n^2 (-1 + η + α η) + n (2 + a α + 2 ks α + (-2 - (2 + a + 2 ks) α + α^2) η) +
             α (α + a (1 + α) (-1 + η) + η - 2 α η + 2 ks (-1 + α + η - α η)))))) /
 (c^2 (1 + c)^2 (-1 + n)^4 - 2 c (1 + c) (1 + a c - (1 + 2 c) kr) (-1 + n)^3
    α +
   (c ((-1 + a) (4 + (3 + a) c) - 2 (a - c + 2 a c) kr + (2 + 3 c) kr^2) - (1 + c)^2 kr ks)
    (-1 + n)^2 α^2 -
   2 ((-1 + a) c (-1 + a - kr) + (1 + c) kr ks) (-1 + n) α^3 -
   kr ks α^4)
```

Reciprocal signatory effort

In[151]:=
```
qrs[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
 (b (-1 + n) (c (1 + c)^2 (kr + ks) (-1 + n)^3 - (1 + c) (-1 + n)^2
          (-ks + (kr + ks) (ks + c (-3 - kr + ks)) + a (-1 + c (kr + ks)) + n + c n) α -
         (-1 + n) (a^2 + ks (-2 + kr + 2 ks) + a (-2 - kr + ks + c (-1 + 2 kr + 2 ks - n) - n) +
            (2 + kr) n + c (ks (-3 + 2 ks) + 2 n + kr (-2 - 2 kr + n))) α^2 -
         (a^2 - ks + ks^2 + n + kr n - a (1 + kr - ks + n)) α^3) +
     α (-kr ks α (-1 + n + α)^2 (-1 + η) + c^3 (-1 + n)^3 (1 - kr + (-2 + kr + n) η) + c^2 (-1 + n)^2
         (-2 + 2 kr + 2 n - 2 kr n + 2 α - 2 a α + a kr α - kr^2 α + kr ks α + 2 (-1 + n) (-2 + kr + n) η -
           (4 + kr - 2 n + a (-3 + kr + n) - kr (kr - ks + n)) α η) + c (-1 + n) (kr^2 (-1 + n) α
            (-1 + η) + (-1 + n + α - a α) (-1 + n + α - a α + (2 + (-2 + a) α + n (-3 + n + α)) η) +
           kr (-1 + η + n^2 (-1 + η + α η) + n (2 + a α + 2 ks α + (-2 - (2 + a + 2 ks) α + α^2) η) +
              α (α + a (1 + α) (-1 + η) + η - 2 α η + 2 ks (-1 + α + η - α η)))))) /
  (c^2 (1 + c)^2 (-1 + n)^4 - 2 c (1 + c) (1 + a c - (1 + 2 c) kr) (-1 + n)^3 α +
    (c ((-1 + a) (4 + (3 + a) c) - 2 (a - c + 2 a c) kr + (2 + 3 c) kr^2) - (1 + c)^2 kr ks)
     (-1 + n)^2 α^2 -
    2 ((-1 + a) c (-1 + a - kr) + (1 + c) kr ks) (-1 + n) α^3 - kr ks α^4)
```

Reciprocal signatory effort without b

```
In[152]:= qrsnob[c_, n_, a_, ks_, kr_, α_, η_] :=
  (c / n (-1 + n) (c (1 + c)^2 (kr + ks) (-1 + n)^3 - (1 + c) (-1 + n)^2
          (-ks + (kr + ks) (ks + c (-3 - kr + ks)) + a (-1 + c (kr + ks)) + n + c n) α -
         (-1 + n) (a^2 + ks (-2 + kr + 2 ks) + a (-2 - kr + ks + c (-1 + 2 kr + 2 ks - n) - n) +
           (2 + kr) n + c (ks (-3 + 2 ks) + 2 n + kr (-2 - 2 kr + n))) α^2 -
         (a^2 - ks + ks^2 + n + kr n - a (1 + kr - ks + n)) α^3) +
      α (-kr ks α (-1 + n + α)^2 (-1 + η) + c^3 (-1 + n)^3 (1 - kr + (-2 + kr + n) η) + c^2 (-1 + n)^2
          (-2 + 2 kr + 2 n - 2 kr n + 2 α - 2 a α + a kr α - kr^2 α + kr ks α + 2 (-1 + n) (-2 + kr + n) η -
           (4 + kr - 2 n + a (-3 + kr + n) - kr (kr - ks + n)) α η) + c (-1 + n) (kr^2 (-1 + n) α
            (-1 + η) + (-1 + n + α - a α) (-1 + n + α - a α + (2 + (-2 + a) α + n (-3 + n + α)) η) +
          kr (-1 + η + n^2 (-1 + η + α η) + n (2 + a α + 2 ks α + (-2 - (2 + a + 2 ks) α + α^2) η) +
             α (α + a (1 + α) (-1 + η) + η - 2 α η + 2 ks (-1 + α + η - α η)))))) /
   (c^2 (1 + c)^2 (-1 + n)^4 - 2 c (1 + c) (1 + a c - (1 + 2 c) kr) (-1 + n)^3 α +
     (c ((-1 + a) (4 + (3 + a) c) - 2 (a - c + 2 a c) kr + (2 + 3 c) kr^2) - (1 + c)^2 kr ks)
      (-1 + n)^2 α^2 -
    2 ((-1 + a) c (-1 + a - kr) + (1 + c) kr ks) (-1 + n) α^3 - kr ks α^4)
```

We can substitute signatory efforts into qrns (Reciprocal non-signatory effort)

```
In[153]:= qrnsf[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
  FullSimplify[qrns[b, c, n, a, ks, kr, α, η, ((1 - n) ((1 + c) (-1 + n) + α)
       (b (c^3 (kr + ks) (-1 + n)^3 - kr (a + ks - n) α^2 (-1 + n + α) + c^2 (kr + ks) (-1 + n)^2
              (-1 + n - (1 + a - 2 kr) α) + c (-1 + n) α ((-2 + kr) (kr + ks) (-1 + n) +
               2 (-1 + a - kr) (kr + ks) α + kr n α)) + c kr α (c^2 (-1 + n)^2 (-1 + η) -
            c (-1 + n) (-1 + n - a α + 2 kr α + η - n η + (-1 + a - 2 kr + n) α η) +
            α (-1 + n + α - a α + kr (-1 + n) (-1 + η) + a α η - n (-1 + n + α) η)))) /
     (c (-1 + n) (-kr ks α^2 ((1 + c) (-1 + n) + α) + c kr ks (-1 + n) α^2 ((1 + c) (-1 + n) + α) +
          kr ks α^2 ((1 + c) (-1 + n) + α) (n + α) - c (1 - n) (-(1 + c) (-1 + n) + (-1 + a - kr) α)
           (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2))),
    (b (-1 + n) (c (1 + c)^2 (kr + ks) (-1 + n)^3 - (1 + c) (-1 + n)^2
            (-ks + (kr + ks) (ks + c (-3 - kr + ks)) + a (-1 + c (kr + ks)) + n + c n) α -
           (-1 + n) (a^2 + ks (-2 + kr + 2 ks) + a (-2 - kr + ks + c (-1 + 2 kr + 2 ks - n) - n) +
              (2 + kr) n + c (ks (-3 + 2 ks) + 2 n + kr (-2 - 2 kr + n))) α^2 -
           (a^2 - ks + ks^2 + n + kr n - a (1 + kr - ks + n)) α^3) +
        α (-kr ks α (-1 + n + α)^2 (-1 + η) + c^3 (-1 + n)^3 (1 - kr + (-2 + kr + n) η) +
           c^2 (-1 + n)^2 (-2 + 2 kr + 2 n - 2 kr n + 2 α - 2 a α + a kr α - kr^2 α + kr ks α +
              2 (-1 + n) (-2 + kr + n) η - (4 + kr - 2 n + a (-3 + kr + n) - kr (kr - ks + n)) α η) +
           c (-1 + n) (kr^2 (-1 + n) α (-1 + η) + (-1 + n + α - a α)
                (-1 + n + α - a α + (2 + (-2 + a) α + n (-3 + n + α)) η) +
              kr (-1 + η + n^2 (-1 + η + α η) + n (2 + a α + 2 ks α + (-2 - (2 + a + 2 ks) α + α^2) η) +
                 α (α + a (1 + α) (-1 + η) + η - 2 α η + 2 ks (-1 + α + η - α η)))))) /
     (c^2 (1 + c)^2 (-1 + n)^4 - 2 c (1 + c) (1 + a c - (1 + 2 c) kr) (-1 + n)^3 α +
       (c ((-1 + a) (4 + (3 + a) c) - 2 (a - c + 2 a c) kr + (2 + 3 c) kr^2) - (1 + c)^2 kr ks)
        (-1 + n)^2 α^2 - 2 ((-1 + a) c (-1 + a - kr) + (1 + c) kr ks) (-1 + n) α^3 - kr ks α^4)]]
```

Indirect Utility Functions

Under symmetry, we had two stability conditions and two indirect utility functions. Under asymmetry, we have 4 of them.
Self-interested non-signatory

In[154]:=
```
indusns[c_, n_, a_, ks_, kr_, α_, η_] :=
  FullSimplify[(c / n) ((n - a - ks) 1/n + (a - kr) qrnsf[c / n, c, n, a, ks, kr, α, η] + ks *
        qss[c / n, c, n, a, ks, kr, α, η] + kr * qrs[c / n, c, n, a, ks, kr, α, η]) - c/2 (1/n)^2]
```

Reciprocal non-signatory

In[155]:=
```
indurns[c_, n_, a_, ks_, kr_, α_, η_] :=
  (c / n) ((n - a - ks) 1/n + (a - kr) qrnsf[c / n, c, n, a, ks, kr, α, η] +
      ks * qss[c / n, c, n, a, ks, kr, α, η] + kr * qrs[c / n, c, n, a, ks, kr, α, η]) -
   c/2 qrnsf[c / n, c, n, a, ks, kr, α, η]^2 +
   α (1/(n - 1) ((n - a - ks) 1/n + (a - kr - 1) qrnsf[c / n, c, n, a, ks, kr, α, η] +
         ks * qss[c / n, c, n, a, ks, kr, α, η] + kr * qrs[c / n, c, n, a, ks, kr, α, η]) - η)
    (qrnsf[c / n, c, n, a, ks, kr, α, η] + 1 - η)
```

Self-interested signatory

In[156]:=
```
induss[c_, n_, a_, ks_, kr_, α_, η_] :=
  (c / n) ((n - a - ks) 1/n + (a - kr) qrnsf[c / n, c, n, a, ks, kr, α, η] +
      ks * qss[c / n, c, n, a, ks, kr, α, η] + kr * qrs[c / n, c, n, a, ks, kr, α, η]) -
   c/2 (qss[c / n, c, n, a, ks, kr, α, η])^2
```

Reciprocal signatory

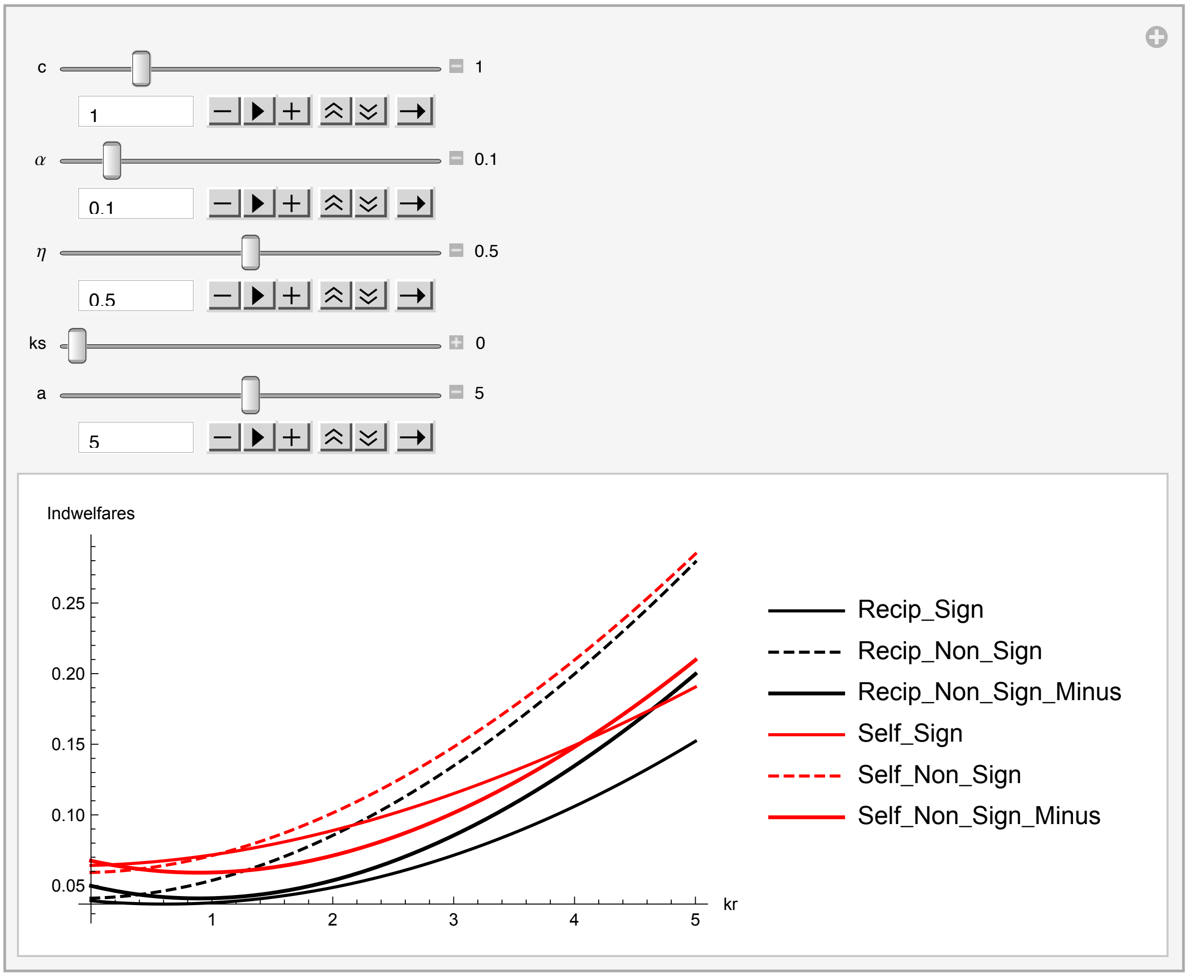
In[157]:=
```
indurs[c_, n_, a_, ks_, kr_, α_, η_] :=
  (c / n) ((n - a - ks) 1/n + (a - kr) qrnsf[c / n, c, n, a, ks, kr, α, η] +
      ks * qss[c / n, c, n, a, ks, kr, α, η] + kr * qrs[c / n, c, n, a, ks, kr, α, η]) -
   c/2 (qrs[c / n, c, n, a, ks, kr, α, η])^2 +
   α (1/(n - 1) ((n - a - ks) 1/n + (a - kr) qrnsf[c / n, c, n, a, ks, kr, α, η] + ks *
          qss[c / n, c, n, a, ks, kr, α, η] + (kr - 1) * qrs[c / n, c, n, a, ks, kr, α, η]) -
      η) (qrs[c / n, c, n, a, ks, kr, α, η] + 1 - η)
```

Stability Conditions (Result 4)

We employ the following figures and equations (below) that we manipulate and numerically analyze the stability conditions for a wide range of parameters. We summarized our results in the paper.

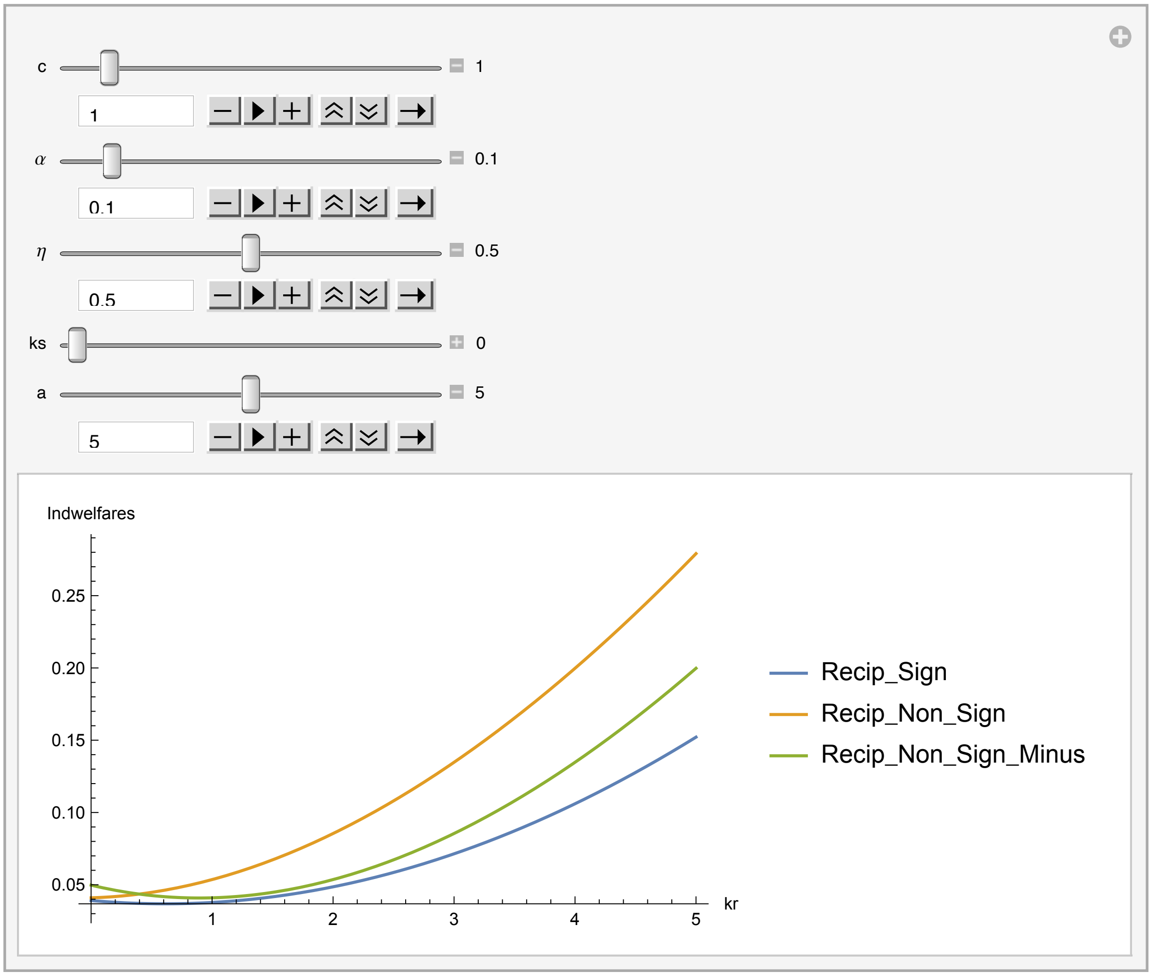
In[158]:=
```
Manipulate[Plot[{indurs[c, 10, a, ks, kr, α, η], indurns[c, 10, a, ks, kr, α, η],
   indurns[c, 10, a, ks, kr - 1, α, η], induss[c, 10, a, ks, kr, α, η],
   indusns[c, 10, a, ks, kr, α, η], indusns[c, 10, a, ks, kr - 1, α, η]}, {kr, 0, 5},
  AxesLabel → {kr, "Indwelfares"}, PlotLegends → {"Recip_Sign", "Recip_Non_Sign",
    "Recip_Non_Sign_Minus", "Self_Sign", "Self_Non_Sign", "Self_Non_Sign_Minus"},
  PlotRange → Full, PlotStyle → {Black, {Black, Dashed}, {Black, Thick},
    Red, {Red, Dashed}, {Red, Thick}}], {c, 0.1, 5, Appearance → "Labeled"},
 {α, 0, 1, Appearance → "Labeled" }, {η, 0, 1, Appearance → "Labeled"},
 {ks, 0, 5, Appearance → "Labeled"},
 {a, 0, 10, Appearance → "Labeled"}]
```

Out[158]=



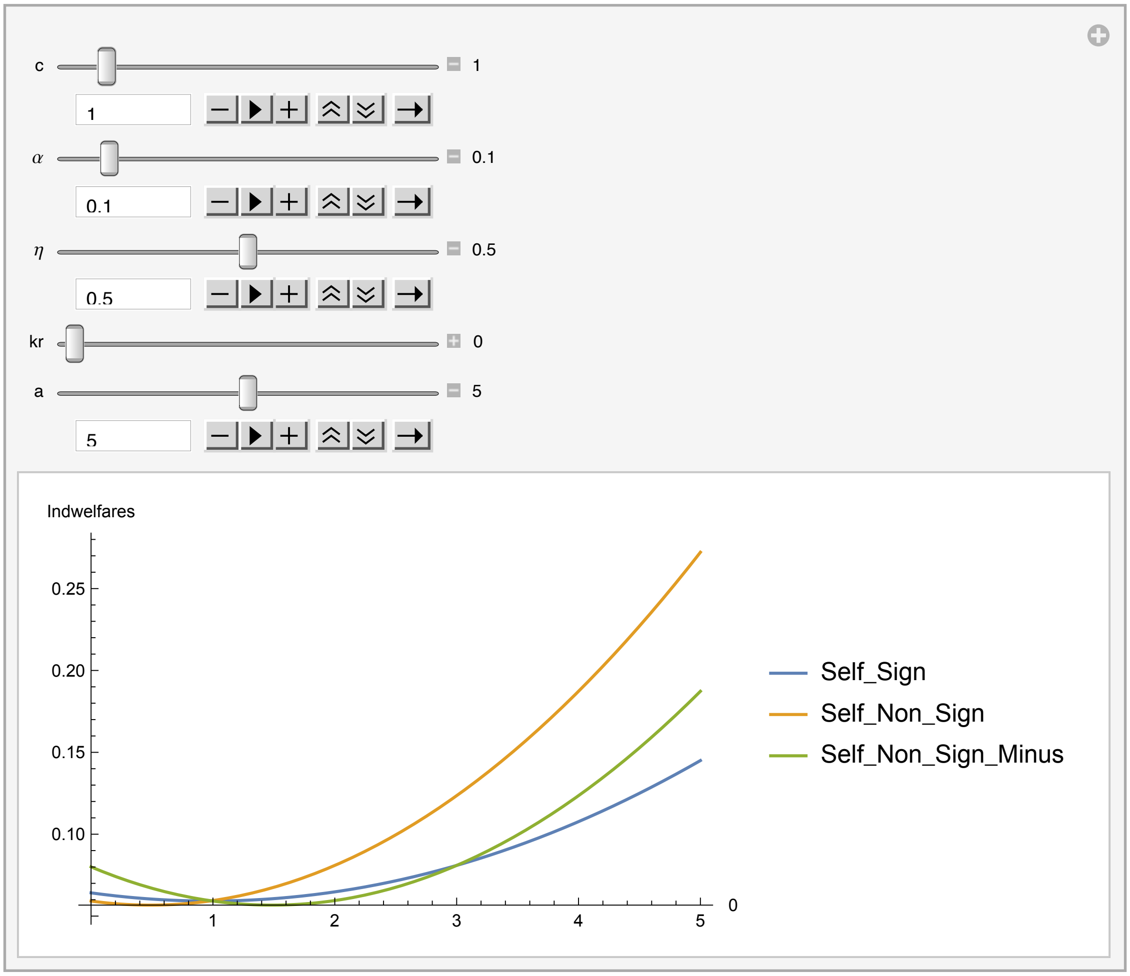
In[159]:=
```
Manipulate[Plot[{indurs[c, 10, a, ks, kr, α, η], indurns[c, 10, a, ks, kr, α, η],
   indurns[c, 10, a, ks, kr - 1, α, η]}, {kr, 0, 5}, AxesLabel → {kr, "Indwelfares"},
  PlotLegends → {"Recip_Sign", "Recip_Non_Sign", "Recip_Non_Sign_Minus"},
  PlotRange → Full ], {c, 0.1, 10, Appearance → "Labeled"},
 {α, 0, 1, Appearance → "Labeled" },  {η, 0, 1, Appearance → "Labeled"},
 {ks, 0, 5, Appearance → "Labeled"}, {a, 0, 10, Appearance → "Labeled"}]
```

Out[159]=



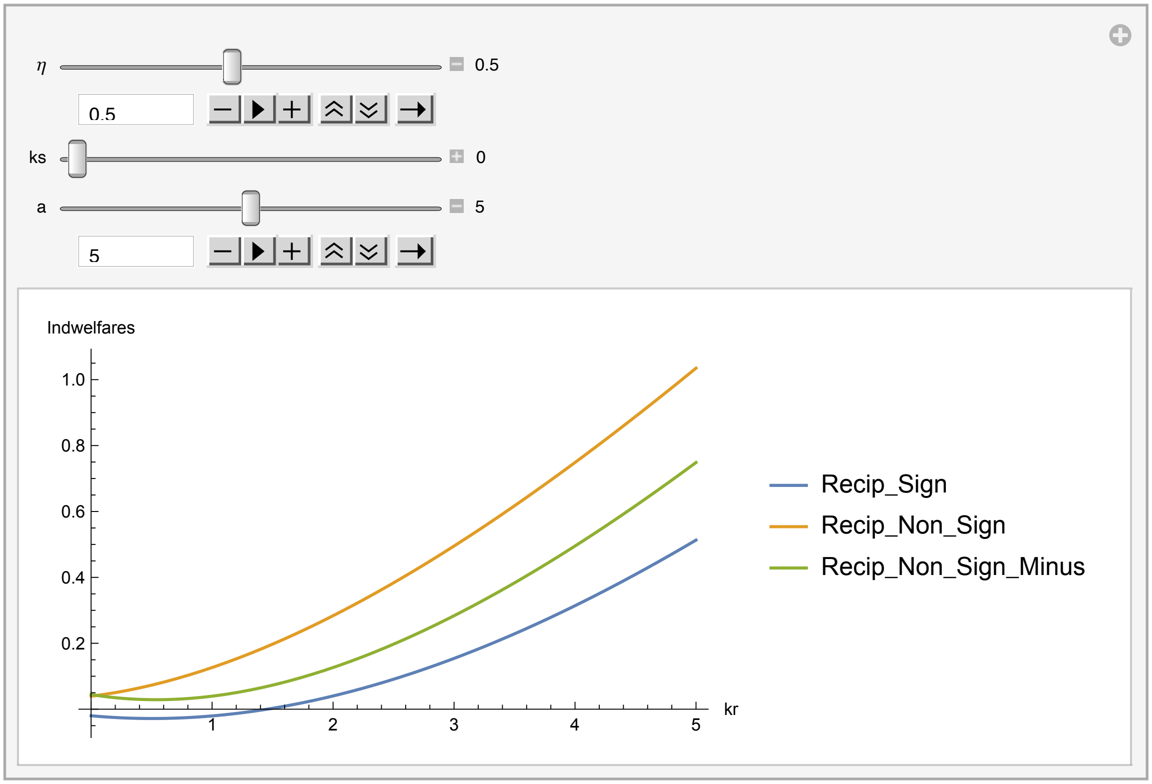
In[160]:=
```
Manipulate[Plot[{induss[c, 10, a, ks, kr, α, η], indusns[c, 10, a, ks, kr, α, η],
   indusns[c, 10, a, ks - 1, kr, α, η]}, {ks, 0, 5}, AxesLabel → {kr, "Indwelfares"},
  PlotLegends → {"Self_Sign", "Self_Non_Sign", "Self_Non_Sign_Minus"},
  PlotRange → Full ], {c, 0.1, 10, Appearance → "Labeled"},
 {α, 0, 1, Appearance → "Labeled" },  {η, 0, 1, Appearance → "Labeled"},
 {kr, 0, 5, Appearance → "Labeled"}, {a, 0, 10, Appearance → "Labeled"}]
```

Out[160]=



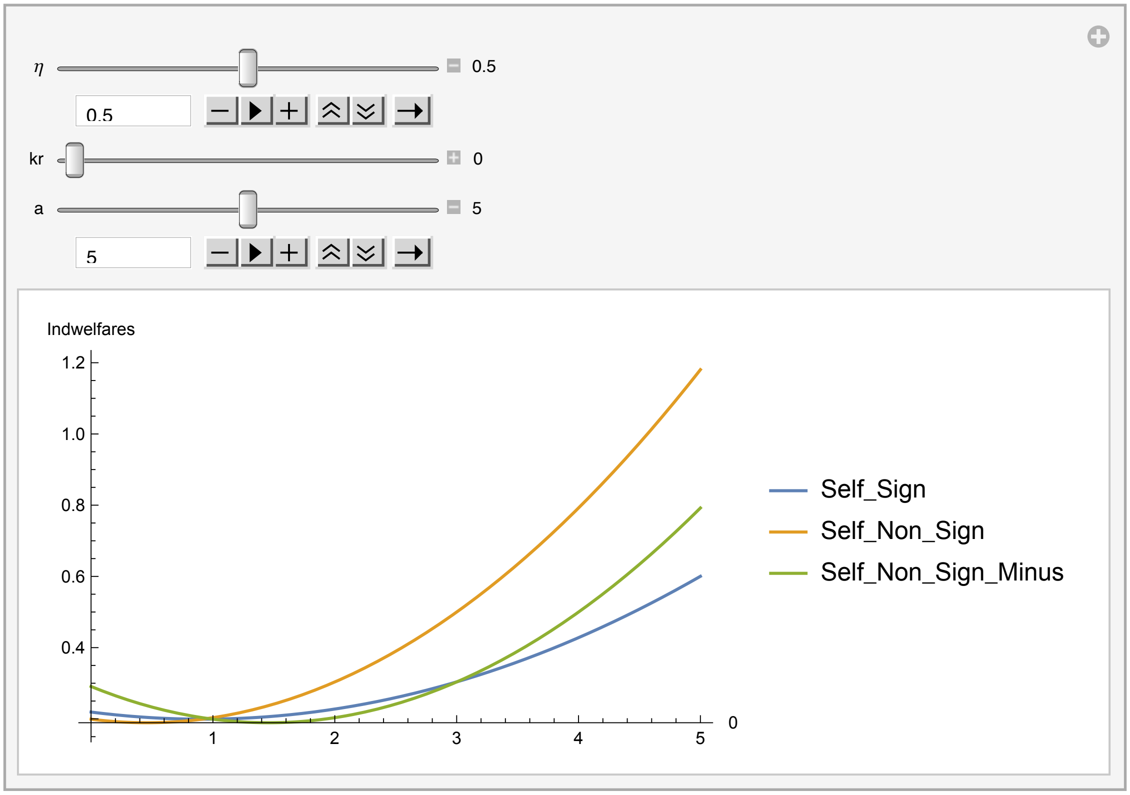
In[161]:=
```
Manipulate[Plot[{indurs[4, 10, a, ks, kr, 0.8, η], indurns[4, 10, a, ks, kr, 0.8, η],
    indurns[4, 10, a, ks, kr - 1, 0.8, η]}, {kr, 0, 5}, AxesLabel → {kr, "Indwelfares"},
  PlotLegends → {"Recip_Sign", "Recip_Non_Sign", "Recip_Non_Sign_Minus"},
  PlotRange → Full ], {η, 0.1, 1, Appearance → "Labeled"},
 {ks, 0, 5, Appearance → "Labeled"}, {a, 0, 10, Appearance → "Labeled"}]
```

Out[161]=



In[162]:=
```
Manipulate[Plot[{induss[4, 10, a, ks, kr, 0.8, η], indusns[4, 10, a, ks, kr, 0.8, η],
   indusns[4, 10, a, ks - 1, kr, 0.8, η]}, {ks, 0, 5}, AxesLabel → {kr, "Indwelfares"},
  PlotLegends → {"Self_Sign", "Self_Non_Sign", "Self_Non_Sign_Minus"},
  PlotRange → Full ], {η, 0, 1, Appearance → "Labeled"},
 {kr, 0, 5, Appearance → "Labeled"}, {a, 0, 10, Appearance → "Labeled"}]
```

Out[162]=



In[163]:=
```
Solve[indurs[4, 10, 5, 3, kr, 0.01, 0.3] ==
  indurs[4, 10, 5, 3, kr - 1, 0.01, 0.3], kr]
```

Solve: Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[163]=
```
{{kr → -4494.15}, {kr → -4494.1}, {kr → -1283.5 - 0.309021 i},
 {kr → -1283.5 + 0.309021 i}, {kr → -2.01849},
 {kr → -0.00313097}, {kr → 9001.51}, {kr → 6.04459 × 10^17}}
```

In[164]:=
```
Solve[induss[4, 10, 5, ks, 0, 0.01, 0.3] == indusns[4, 10, 5, ks - 1, 0, 0.01, 0.3], ks]
```

Out[164]=
```
{{ks → 0.999999}, {ks → 3.}}
```

In[165]:=
```
qssnob[4, 10, 5, 0, 5, 0.95, 0.1]
```

Out[165]=
```
0.335977
```

In[166]:=
```
qrsnob[4, 10, 5, 0, 5, 0.95, 0.1]
```

Out[166]=
```
0.343119
```

Let's start from the case where kr=ks=0 and check each type of country's incentives to become a signatory in a stable coaltion.
Let us initially assume α = 0.2.

```
1. eta = 0.5 and a = 5.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3; stable
Stable coalition: (ks*, kr*) = (3,0)

Result: There can be a stable coalition formed by only self-interested. Coalition
size is 3.

2. eta = 0.2 and a = 5.
ks = 0 -> kr = 3 -> ks = 0; stable
ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3 - > kr = 0; stable
Stable Coalitions: (ks*, kr*) = {(0,3),(3,0)}

Result: There can be multiple stable coalition.

3. eta = 0.8 and a = 5.
ks = 0 -> kr = 0 -> ks = 3 -> kr = 0; stable
Stable Coalition: (ks*, kr*) = (3, 0)

Result: The higher the eta, the less willing the reciprocal countries to join the
coalition.

------
4. eta = 0.5 and a = 8.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3, but there are
no 3 self-interested.

Result: No Stable Coalition

5. eta = 0.2 and a = 8.
ks = 0 -> kr = 3 -> ks = 2 -> kr = 0 - > ks = 3, but there are no 3 self-inter-
ested -> ks = 2
ks = 1 -> kr = 1 -> ks = 2
Stable Coalitions: (ks*, kr*) = (2,0)

Result: Stable coalition size can be 2.

6. eta = 0.8 and a = 8.
ks = 0 -> kr = 0;
Stable Coalition: (2,0)

Result: The higher the eta, the less willing the reciprocal countries to join the
coalition.
-----
7. eta = 0.5 and a = 2.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3, kr = 0; stable
Stable Coalition: (ks*, kr*) = (3,0)

8. eta = 0.2 and a = 2.
ks = 0 -> kr = 3, but there are no 3 reciprocal, kr = 2 -> ks = 1 -> kr =1 -> ks =
2 -> kr = 0 - > ks = 3 -> kr =0; stable
Stable Coalitions: (ks*, kr*) = (3,0)

9. eta = 0.8 and a = 2.
ks = 0 -> kr = 0 -> ks = 3 -> kr = 0; stable. With sufficiently high eta recipro-
cal countries don’t want to join.
Stable Coalitions: (ks*, kr*) = {(3,0)}
```

Let’s check some other parameters:

1a. alpha = 0.5, eta = 0.3 and a = 5.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 0 -> ks = 3 -> kr = 0 - > ks = 3 -> kr = 0; stable
Stable coalition: (ks*, kr*) = (3,0)

2a. alpha = 0.1, eta = 0.3 and a = 5.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3 -> kr =0; stable
Stable coalition: (ks*, kr*) = (3,0)

3a. alpha = 0.01, eta = 0.3 and a = 5.
ks = 0 -> kr = 3 -> ks = 0; stable
ks = 1 -> kr = 2 -> ks = 1; stable
ks = 2 -> kr = 2 -> ks = 1; stable
ks = 3 -> kr = 0 -> ks = 3; stable
Stable Coalition: (ks*, kr*) = {(0,3),(1, 2),(2,1),(3, 0)}

Result: For sufficiently small alpha, self-interested and reciprocal countries can jointly form coalitions.

More Results: Sufficiently high alpha coupled with sufficiently low eta yields a stable coalition form by all reciprocal countries without any self-interested.
Example: (c,n,a,ks, kr,α,η) = (4, 10, 5, 0, 5, 0.95, 0.a)